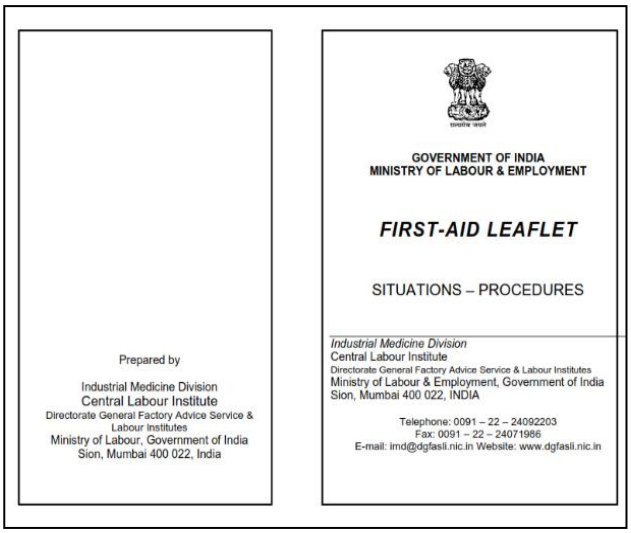
# 8. AUDIT OBSERVATION AND RECOMMANDATIONS

## 8.1 Annexure - A: Areas of Improvement

*(Note- All Pics in recommendation are as reference and for better clarity only)*

| Sr. No. | Observation | Recommendation |
|---|---|---|
| **Legal / General** | | |
| 1. | Record of mandatory Health & Safety Training imparted to workers under Section 111 A (ii) of The Factories Act, 1948 was not evident. | Under Section 111 A (ii) of The Factories Act, 1948, Mandatory Health and safety training on various topics viz; Accident prevention techniques, MSDS, Machine safety, Work permit system, Fire prevention & firefighting, Emergency plan etc. shall be imparted to workers by DISH approved training institute. |
| 2. | Environment, Health & Safety Policy was available and displayed in English only. | As per rule 68-O of the Gujarat Factories Rules, 1963 Policy should be made available in local language and displayed it at conspicuous places. |
| 3. | First aid box available across the factory but copy of FIRST-AID LEAFLET (Situations- Procedures) issued by the Directorate general of Factories Advice Service and labour Institutes, Government of India, Mumbai was not evident in all First Aid Boxes. | As per the Rule 70 (C) (xxiii) of the GFR, 1963, one Copy of first aid leaflet issued by the Directorate general of Factories Advice Service and labour Institutes, Government of India, Mumbai shall be kept in each first aid box.  Prepared by<br>Industrial Medicine Division<br>Central Labour Institute<br>Directorate General Factory Advice Service & Labour Institutes<br>Ministry of Labour, Government of India<br>Sion, Mumbai 400 022, India<br><br>GOVERNMENT OF INDIA<br>MINISTRY OF LABOUR & EMPLOYMENT<br>FIRST-AID LEAFLET<br>SITUATIONS – PROCEDURES<br>Industrial Medicine Division<br>Central Labour Institute<br>Directorate General Factory Advice Service & Labour Institutes<br>Ministry of Labour & Employment, Government of India<br>Sion, Mumbai 400 022, INDIA<br>Telephone: 0091 – 22 – 24092203<br>Fax: 0091 – 22 – 24071986<br>E-mail: imd@dgfasli.nic.in Website: www.dgfasli.nic.in |
| 4. | Confined space entry (to enter any chamber, tank, vat, pit, pipe, flue, and reactor) was done by without competent person authorization. | As per Section 36 of The Factories Act, 1948, any confined space entry shall be done by taken certificate from competent person approved by DISH [Directorate of Industrial safety & Health]. |
| 5. | Display of the confined space on all the manhole covers inside the plant was not evident. | As per Rule 102, Schedule 19, Part II, Para (18) (1) (a) of The Gujarat Factories Rules, 1963, Confined pace entry to be identified and to be displayed.<br>To be mentioned suitably for all the manholes of various equipment in the plant |

| Sr. No. | Observation | Recommendation |
|---|---|---|
| | | |
| 6. | Visitor safety induction film was not available. | Visitor safety induction film to be made available near security gate. |
| 7. | Colour coding system for pipeline (utilities and chemicals) was not available. | Colour coding system for pipeline should be maintained as per IS 2379 & colour code to be displayed the board in the plant.<br><br>**Basic identification colours**<br>Pipe Contents Name Reference / Colour / Colour<br>Water / Green<br>Steam / Silver Grey<br>Oils (mineral, vegetable or animal) Combustible liquids / Brown<br>Gases in either gas or liquid phase - except air / Yellow Ochre<br>Acids / Alkalis / Violet<br>Air / Light Blue<br>Waste effluents / Black<br>Electrical Services & Ventilation Ducts / Orange<br>**Safety Colour**<br>Safety Reference / Colour / Colour<br>Fire / Red<br>Water from a public supply / Auxiliary blue<br>Water from any other source / Flint grey<br>Warning / Yellow |
| 8. | Cautionary Notice for Acids & Alkalis near HCl storage and handling area was missing. | Precautionary notice as per Paragraph 6 of Schedule XII (Manipulation of Acids or Alkalis) under Rule 102 The Gujarat Factories Rules, 1963;<br>• Cautionary Notice in following manner shall be displayed in language understood by the majority of workers employed.<br>• If any worker is illiterate, effective steps shall be taken to explain carefully to him contents of the notice so affixed: -<br><br>**CAUTIONARY NOTICE**<br>**DANGER**<br>**Acids and Alkalis cause severe burns and vapours thereof may be extremely hazardous. In case of contact, immediately flood the part effected with plenty of water for at least 15 minutes.** |
| 9. | SOP/Do's & Don'ts not displayed at all operation areas and Chemical storage area. | SOP/Do's & Don'ts to be displayed at all operation areas and Chemical storage area. |

| Sr. No. | Observation | Recommendation |
|---|---|---|
| | | |
| 10. | Record of undertaking taken from workers for giving training on cautionary notice and instruction not evident. | As per the Para – 4 of Part – II of Schedule XIX under Rule 102 of The Gujarat Factories Rules, 1963 An undertaking from the workers shall be obtained within 1 month of their employment and for old workers employed within one month of coming in to operation of these rules, to the effect that they have read the contents of the cautionary notices and instruments, understood by them and would abide by them. |
| 11. | MOC (Management of change) Procedure was not evident for all changes/ process. | MOC (Management of Change) Procedure should be established/reviewed, implemented and record to be maintained. |
| 12. | Work place monitoring records was not evident. | Work place monitoring to be carried out at regular interval and records to be maintained in form-37. |
| 13. | Over all housekeeping and illumination was not to the required level for smooth movement for safe operation. | It is recommending for improving housekeeping of operation all area, as this is the first cause for any incident. Different level of illumination (lux level) is required as per the activity. Plan for illumination survey as per the standard by a competent vendor and if, less than standard lux level provide additional light fitting. |
| 14. | Piping color-coding chart display found in operational area but not followed as most of the pipelines colour was peeled off. | After rectification of pipelines, repainting pipeline as per piping colour code to avoid mal charging or isolation chemical |
| 15. | Flow direction arrows marking not found for all pipelines. | Mark flow direction arrow on all pipelines after repaints of pipelines, which will help for quick and correct opening or closing of valve during emergency. |
| 16. | Hazard area classification display not found in that area. | Plan for hazard area classification & display zone name in operational area. |
| 17. | HDPE and PPRC pipeline used for transferring chemical and utilities but they were bend & hanging as they were not supported properly. This may lead leaks and incident. | Consult HDPE / PPRC pipe supplier and provide support as per their guideline. |

| Sr. No. | Observation | Recommendation |
|---|---|---|
| 18. | Only 04 windsocks provided in complex and not visible from all locations & may misguide for escape during emergency. | Make survey that windsock is visible from all locations. If short fall, plan to fix new windsocks at an appropriate location. |
| 19. | Plant was of wide area with high working staff strength, but emergency escape route with arrows & symbols were not found / very less in operation area. | Paint or fix sticker for Emergency escape route with arrow & symbols in fluorescent light, which can be easily seen during power failure and from a working area. It is also advisable to provide at junction where there are more than one way. |
| 20. | a) Reactor agitator and pump coupling covered with guard but most of them was not as per the standard, which are in hose made with wide opening, loose / not rigid, not covering all rotating parts and not of proper MOC proper MOC (Few agitator coupling covered with polymer net).<br>b) At few places common canopy provided for motor and pump coupling, on dislocation or inter change of canopy, coupling got expose and result human body injury. | a) This is a statutory requirement to cover all rotating parts to prevent from body injury. It is recommended to provide rigid guard as per the standard to cover all rotating parts and anchor to ensure that it will be in position all the times.<br>b) Common canopy of motor & coupling not fulfil the purpose of coupling guard, so provide separate guard for coupling. |
| 21. | Nitrogen line provided to reactor / tank for purging / blanketing / material transferring but there was no NRV in nitrogen line considering nitrogen pressure was more than reactor / tank. Possibility to pass of tank vapor to nitrogen header if valve was passing / not close properly and drop in nitrogen header pressure at the same time. | It is recommended to provide NRV in nitrogen line near each reactor / tank. |
| **Plant – 1** | | |
| 22. | Reactor floor had opening on floor and having trip and fall hazard. | Cover opening by grating or provide sufficient height toe guard at tank periphery. |
| 23. | Instrument Y tree header drain and unused tapings were isolated by closing valve but not plugged. Anyone can open this valve for any general-purpose use, drop in IA pressure & results unwanted operation of pneumatic control valve. | Plug instrument Y tree drain & unused tapings on above closing of valve. |
| 24. | Pipelines without piping colour code, without flow direction arrow & | It needs for painting all pipelines as per colour code, mark flow direction and provide support to HDPE lines. |

| Sr. No. | Observation | Recommendation |
|---|---|---|
| | HDPE pipes without proper supports. | |
| 25. | Fire extinguishers not at marked location of FE/07, 10, 14 &… | Put FE form stand by stock before old FE. |
| 26. | R-111 had no agitator coupling guard | Provide coupling gaud as per standard. |
| **Plant – 2** | | |
| 27. | Ground floor had very low illumination. | Plan for illumination survey. |
| 28. | R-202 (SS) Hexane handling line had no bonding / flange jump and free flow charging from top of the reactor. There will be accumulation of generated static charge | Provide bonding on Hexane pipe line and charge in reactor with J type pipe. |
| 29. | Sodium cyanide handling area display with Do and Don't but MSDS extract with pictorial use of PPWE not displayed. On line breathing set use while charging NaCN handling but it was dirty, no pressure gauge and moisture / dust filter and no guide line for use of online BA set. | Display pictorial view for use of on line BA set and required PPE. Keep the safety items in ready to use condition all the time. |
| 30. | Marking not done for emergency light, so it was difficult to judge whether it was normal light or emergency light and if not getting on during normal power failure, people may assume this was normal light fitting. | Tag emergency light by fixing sticker as "Emergency Light" or paint red colour on fitting. |
| **Plant – 3** | | |
| 31. | Hand rail was missing on all turning platform of staircase and had sufficient gap for fall hazard for man and material. | Rectify hand rail and steps for protection of fall hazard. |
| 32. | Cooling water main supply and return line highly corroded | Plan for painting of pipelines. |
| 33. | R-304 reactor piping flanged corroded and no agitator coupling guard. | Rectify the pipeline and provide coupling guard |
| 34. | Nitrogen line provided in R-314 but no NRV in nitrogen line. | Provide NRV in nitrogen line. |
| 35. | Bromine handled in glass pot and pipes but there was no protection of them from impact load, result damages and spillage of highly toxic chemical. | Plan for protection of glass vessel / pipeline or restrict un authorised movement and display instruction / guide line for working in this area. |

| Sr. No. | Observation | Recommendation |
|---|---|---|
| 36. | Acid and caustic handled in this area but safety shower had no any cautionary notice. | Display notice as below near safety shower.<br>**CAUTIONARY NOTICE**<br>Acid and Alkalis cause severe burns & vapor thereof may be extremely hazardous. In case of contact, immediately flood the effected parts with plenty of water for at least 15 minutes. |
| 37. | PPE matrix displayed in plant but awareness of PPE matrix and use of PPE was not to the required level. | Need training to working staff and instruct to use required PPE as per matrix before start of a particular job. |
| **Chlorine Handling** | | |
| 38. | PESO license displayed but it was showing license number and its validity date. | Display license with full details mentioning number of tonner. |
| 39. | Chorine detector provided at more than 1.5 meter height from floor level in chorine gas cylinder area, this may not detect at small leak or may take time or detect at heavy leakage, as chorine is heavier than air. | Provide chlorine detector at lower height for better performance. (Refer chorine detector supplier guideline for installation / exact height from floor level). |
| 40. | Chlorine toner handling EOT:<br>EOT details not displaced in nearby area. | This is statutory for EOT chain blocks to display equipment number, SWL, Tested date & Next test due date on EOT or at a nearby visible location. |
| 41. | Filled tonner kept with very small size stopper, may has possibility for roll out. | Use only standard size stopper like wheel choke for storing / stacking of filled and empty tonners. |
| 42. | Empty tonners were stacked with both valve cap but without fixing valve protecting hood, which require to protect valve from damage during transport and internal handling as empty toner still had some balance chlorine & this may leak due to damage. | It is recommended to Fix valve protection hood before lifting out empty tonner from its position. |
| **Bromine Storage (3 service + 1 s/b Bullet)** | | |
| 43. | Bromine bullet tank top platform had wide opening, this having trip and fall hazard. | Cover opening by grating or plate or provide sufficient height toe guard at bullet periphery. |
| 44. | Unused pipeline tapping on standby bullet isolated by closing valve but not blinded so, possibility for bromine leak. | Isolation of unused branch tapping by simple closing of valve is not reliable in highly toxic and flammable material handling system. Valve is supposed to pass due to frequent operation or not closed properly. All to be blinded or Plugged with a competent MOC and specified rating. |
| **Plant – 5 & 7** | | |
| 45. | R-502 MSGL solvent / Chemical charging line had no bonding and free flow from top in reactor. | Provide bonding in solvent lines and extend solvent line dip at bottom and confirm that reactor had |

| Sr. No. | Observation | Recommendation |
|---|---|---|
| | | tantalum plug if not, develop facility for release of generated static charge. |
| 46. | R-504 toluene transfer line had no boding | Provide bonding. |
| 47. | R-504 and other reactor agitator coupling had of poly net guard. | Provide coupling guard as per guideline. |
| 48. | R-519 reactor and nearby pipelines insulation was provided but cladding not fixed, so water /chemical may ingress in between insulation and pipe outside, accumulated water deteriorate insulation quality and leads metal corrosion called as corrosion under insulation (CUI). | Rectify damaged cladding and provide at missing locations & make practice to apply cladding just on completion of insulation. |
| 49. | There was no MCP, Fire Call point and escape route arrow. | It needs to review for providing MCP, Fire Call point and escape route arrow at an appropriate locations. |
| **Plant – 6** | | |
| 50. | Heavy storage of scrap at entrance area and reduce plant looks | Needs shift scarp at a designated location. |
| 51. | Near R-616 steam line without insulation which had human burn injury. | Provide insulation on line or if not feasible plan for personal protection which requires, where surface temperature is more than 5o degree centigrade. |
| 52. | Fire extinguisher blocked by chemical filled open drums, so difficult to access FE and fume was coming out from open drums. This may be fire and health hazard. | Keep the firefighting and safety items approach without any obstruction and close the open chemical drums. |
| 53. | Low illumination at ground floor. | Plan for illumination survey. |
| **FWPH** | | |
| 54. | Fire Flow Head Start Pr. Stop Pr.<br>Pump $M^3/Hr$ Meter $Kg/Cm^2$ $Kg/Cm^2$<br>Jokey 10.8 70 6.0 8.0<br>M.Ele. 273 70 5.0 Manual<br>M. Die 273 70 Manual Manual<br><br>Required fire water system details were not found near FWPH entry. | It is recommended to display following details near FWPH entrance.<br>• Each pumps' Capacity in $M^3/Hr$, Head in meter, start pressure in $Kg/Cm^2$ and stop pressure in $Kg/Cm^2$.<br>• Fire water reserve capacity in $M^3$.<br>• Hydrant number & Monitor number.<br>All these details required to check present firewater facility against the requirement. |
| 55. | a) Main electrical fire water pump had common canopy for motor and coupling, but coupling not covered fully.<br>b) Electrical panel without rubber mat | a) Provide separate guard on coupling as per the standard.<br>b) Provide IS mark rubber mat in front of panel. |
| **11 KV / 440 V – Yard** | | |
| 56. | Fencing height was less than 1.8 meter. | Fencing height to be 1.8 meter. |

| Sr. No. | Observation | Recommendation |
|---|---|---|
| **PCC / MCC – 1** | | |
| 57. | Display of "Unauthorized entry Restricted" & "List of authorized persons" not found. | Display board for "Unauthorized Entry Restricted." & "List of authorized persons" at entrance. |
| 58. | Opening in wall for cables not sealed and without fire retardant paint, so fire may travel to other room / portion if occur in any one side due to short circuit or over load in cable. | Seal cable entry & exit wall opening, and apply fire retardant paint on cable approximate for one meter on both sides, so cable fire can be stopped to go in next nearby compartment. |
| 59. | MCC room housekeeping was no to the required level. | Remove unwanted material and improve housekeeping. |
| **PCC / MCC – 2 (First Floor)** | | |
| 60. | Display of "Unauthorized entry Restricted" & "List of authorized persons" not found. | Display board for "Unauthorized Entry Restricted." & "List of authorized persons" at entrance. |
| 61. | PCC room had HVAC & Exhaust fan still room was hot, Cooling to panel done by 04 number of portable pedestals. | It needs to study the ventilation system capacity and performance of exhaust fan capacity and number. |
| 62. | Pedestal fans were without additional fine screens on fan blade guard, this had human finger injury. | Make study to replace pedestal fan by high-capacity exhaust fan for room cooling, if not feasible, provide fine screen on blade guard and protect from finger injury. |
| 63. | Emergency gate provided but exit arrow not found. | Mark arrow showing direction of emergency exit. |
| 64. | First floor staircase had damaged step & railing this had fall hazard. | Rectify the staircase step and hand railing. |
| **Common observations for all tank farm** | | |
| 65. | There was no evidence for use of wheel chokes while parking of tanker for loading / unloading in all loading / unloading points. In some cases, front portion of tanker lifted / kept on stand while tanker unloading to empty out tanker fully, and parked by using hand brake. Tanker goes back if hand break failed. This result-unwanted movement of tanker, detach of loading / unloading hose and spillage of chemicals. | It is recommended for using minimum two wheels chokes in opposite directions while doing tanker loading or unloading activity. |
| 66. | At few tank farm, tanker activity done on road and may obstruct / restrict movement emergency vehicles like fire tender and ambulance. | Survey Tank farm tanker activity, mark area for tanker location for loading / unloading activity, keeping clear path for movement of emergency vehicle like fire tender and ambulance. |
| **IB TF – 02 Tanks** | | |

| Sr. No. | Observation | Recommendation |
|---|---|---|
| 67. | IB tank top platform had wide opening, this have trip and fall hazard. | Cover opening by grating or plate or provide sufficient height toe guard at tank periphery. |
| **TF of Plant – 2** | | |
| 68. | A tank detail was not available as painting was going on. | Display following details on each tank: Tank number, Name of chemical or material code, Maximum storing capacity & NFPA diagram at such a location, height, letter size, so which can be readable from a road. |
| 69. | Dike drain valve found open, this defeats purpose of dike. | Drain valve to be kept closed all the time except draining for rainy water. |
| **TF of Plant – 1** | | |
| 70. | Tanks were in different dike but few dike walls punctured to take pipes from one dike to other. Possibility of inter mixing of chemicals which may not be compatible. | Seal opening in tank dike wall. |
| **FG W/H – 1** | | |
| 71. | FG loaded in truck from elevated platform at entrance, but this open space had fall hazard if there was parked vehicle. | Plan for detachable barricading / hand rail which to be fixed when there was no loading activity. |
| 72. | Fork lift (FL) Driver was not wearing seat belt. | It is recommending to Develop fork lift check list and to be filled daily before start of FL for healthiness, if any point not comply, concession to be taken from concerned authority. |
| **RM W/H** | | |
| 73. | There was no identified inspection zone to test incoming material, possibility to issue of un tested material to user. | Identify area, mark and display as material under inspection. Shift the material to relevant approved or rejected material area as per test report. |
| **Utility – 1** | | |
| 74. | VC-2:<br>$CaCl_2$ circulations pump without coupling guard. | Provide guard on coupling. |
| 75. | Nitrogen PSA:<br>Huge amount of nitrogen handled in PSA plant. People were not fully aware for Nitrogen hazard and precaution to be taken while handling / leak of nitrogen. | It is recommended to give training to working staff for nitrogen handling, display nitrogen Do & Don't guideline. Asphyxia hazard warning signage and MSDS main critical / few important parameters with main hazard, symbolic use of relevant PPE & NFPA diagram in such a language, so maximum staff can understand. |
| **Utility – 2** | | |
| 76. | Cooling water: Approach is difficult due to water logging and stored scrap near CW pumps and both CW pump were without coupling guard. | It needs corrective action on regular basis. Provide guard on CW pump coupling. |

| Sr. No. | Observation | Recommendation |
|---|---|---|
| **QC** | | |
| 77. | Chemical MSDS hard copy kept in laboratory but there was no display of chemical comparability chart which were used in laboratory. | It is recommended to study chemical compatibility for all chemicals stored in laboratory, prepare chart, display & follow the same for storing of chemicals. |
| 78. | Fuming chamber performance was poor as its front door not closing properly, hole in fuming chamber and without connection of exhaust blower, so all fume in working area. | Rectify the fuming chamber or plan for new one with all features. |
| **ETP / MEE** | | |
| 79. | Housekeeping was found poor. | Housekeeping needs improvement. |
| 80. | Illumination was found poor in hazardous waste storage area. | Illumination study to be carried out and accordingly illumination to be improved. |
| 81. | Near MEE area, water logging observed and also floor was slippery. | To be reviewed. |

# 1. Occupational Health & Safety Audit as per IS 14489: 2018

## 9.1 ANNEXURE – B: Examination of Records (IS 14489: 2018)

Observations including nonconformities and good practices by the auditee factory are given below in Annexure B to C. Necessary recommendations / suggestions or improvement measures are also suggested in the last column.

| Sr. No. | List Of Records / Documents | Pertaining Legal Requirement | Current Status / Observation |
|---|---|---|---|
| 1. | OH & S policy | The GFR- Rule 12-(C), 68- O. | OH & S policy available in the Organisation. |
| 2. | Safety organization chart | The FA 1948 – Section 40- B. | Safety Organisation Chart available in the Organisation. |
| 3. | Training records on safety fire and first-aid | The FA 1948 - 7 (A), 111 (A)<br>The GFR- Rule GFR – 12 (C) 4(g). | Training records on safety fire and first-aid maintained and available in organization. |
| 4. | Record of plant safety inspections | The GFR- Rule 112. | Record of safety inspections maintained and available in organization (Monthly safety inspection carried out by safety officer). |
| 5. | Accident investigation reports | The GFR- Rule 103, 111. | Accident investigation reports was available with EHS department. |
| 6. | Accidents, dangerous occurrences and near miss incidents - statistics and analysis | The GFR- Rule 103, 111. | Accidents, dangerous occurrences and near miss incidents - statistics and analysis were available with concern department. |
| 7. | Record of tests and examinations of equipment and structures as per statutes | The GFR- Rule 58, 59, 60,102. | Form No. – 9, 10, 11 and stability certificate available. |
| 8. | Standard Operating Procedures (SOP) for various operations | --- | Standard operating procedures were available for various operations. |
| 9. | Record of work permits; | The GFR- Rule 102 Chemical works sch. 19. | Record of work permits were available.<br>**List of work permits:**<br>1. Hot work<br>2. Confined Space<br>3. LOTO<br>4. Cold work<br>5. Line opening<br>6. Height Work<br>7. Excavation |

| Sr. No. | List Of Records / Documents | Pertaining Legal Requirement | Current Status / Observation |
|---|---|---|---|
| | | | 8. Hazardous chemical loading / unloading |
| 10. | Record of work environment monitoring (Flammable, toxic and explosive substances) | The GFR- Rule 68 J and 12 – B<br>The FA 1948 Section 7-A (2) (E). | Record of work environment monitoring (Flammable, toxic and explosive substances) was available. |
| 11. | Maintenance, testing and calibration records of fire detection and firefighting equipment | The GFR- Rule – 66A. | Maintenance, testing and calibration records of fire detection and firefighting equipment was available. |
| 12. | Medical records of employees | The GFR- Rule 68 R. | Medical records of employees (Form No. 32, 33) were available. |
| 13. | Records of industrial hygiene surveys (noise, ventilation, illumination, dust etc.) | The GFR- Rule - Rule 12-B<br>The FA 1948 - Section 7A (2)(E). | Records of industrial hygiene surveys (noise, dust, illumination, etc.) was available. |
| 14. | Material Safety Data sheets | The GFR- Rule 68 J, MSIHC Rule 1989, Rule 17, Schedule 9. | MSDS of All chemicals, RM, Intermediates and Product were available. |
| 15. | On-site emergency plans and record of Mock Drills | CA EPPR 1996, The GFR- Rule 68 J, The FA 1948 Schedule – 8-A. | On-site emergency plan was available.<br>Last updation: August 2023<br>Record of Mock Drills was available.<br>Last mock drill carried out: - 28/03/2024 |
| 16. | Records of storage of hazardous solid waste and its disposal | The GFR- Rule - 18, The FA 1948 section 12. | Records of storage of hazardous solid waste and its disposal was available. |
| 17. | Records of gaseous emissions and effluent discharges to the environment | The GFR- Rule - 18, The FA 1948 section 12. | Records of gaseous emissions and effluent discharges to the environment was available at Env. Department.<br>(Monthly env. Monitoring done by M/s. Unistar Lab) |
| 18. | Housekeeping inspection records | The GFR – Rule 16, 17, 17 A, The FA 1948 section 11. | Housekeeping inspection records was available. |
| 19. | Minutes of safety committee meetings | The GFR- Rule – 68 F, Y. | Minutes of safety committee meetings was available.<br>Quterly meeting done.<br>Total – 28 members<br>Last meeting held on 18/01/2024<br>Chairman: - Mr. D. N. Rai<br>Secretary: - Mr. Ashish Joshi |

| Sr. No. | List Of Records / Documents | Pertaining Legal Requirement | Current Status / Observation |
|---|---|---|---|
| 20. | Statutory licenses and approvals | The GFR- Rule -3. | Statutory licenses and approvals were available. |
| 21. | Records of any modifications carried out in plant or process | The GFR- Rule -3,7. | Records of any modifications carried out in plant or process was available informally. |
| 22. | Maintenance procedure and records | --- | Preventive Maintenance procedure and records were available. |
| 23. | Instrumentation and equipment calibration and testing records | The FA 1948, Section 36. | Instrumentation and equipment calibration and testing records available at technical department |
| 24. | Planned shutdown maintenance procedures | --- | Procedure for planned shut down was available. |
| 25. | In service inspection manuals, records including that of material handling | --- | All material handling equipment service inspection records, manuals available. |
| 26. | OH & S budget | The GFR- Rule 12 C & 68-0. | OH & S budget available. |
| 27. | Inspection books and other statutory records; | The GFR- Rule – 112. | Inspection books and other statutory records were available. |
| 28. | Records of previous audits and safety analysis; | The GFR- Rule 68 J | Records of previous audit was available & previous audit compliance report also available. |
| 29. | Procedures for safe transportation of hazardous substances; | MV Act. | Procedures for safe transportation of hazardous substances was available. |
| 30. | Calibration records; | --- | Instrumentation and equipment calibration and testing records available at technical department. |
| 31. | Records for breakdown of plants during the process of manufacture; | --- | Records for breakdown of plants during the process of manufacture was available at technical department. |
| 32. | Records for waste material generated and their disposal; | --- | Records for waste material generated and their disposal record was available. |
| 33. | SOP for disposal of waste materials; and | --- | SOP for disposal of waste materials was available. |
| 34. | Records for issue of PPE items to the personnel working in process building. | --- | Records for issue of PPE items to the personnel working in process building was available at safety department. |

## 9.2 ANNEXURE – C: General Points (IS 14489: 2018)

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | **C-1 OH & S Management** | | |
| | **C-1.1 OH & S Policy (Rule 12-C And / Or 68-O of The GFR 1963)** | | |
| 1. | Does the organization have OH & S Policy? | Organization have EH&S Policy. | --- |
| 2. | Who has signed the OH & S policy? | EH&S Policy was signed by - Mr. D. N. Rai | --- |
| 3. | Whether the OH & S policy is per guidelines of the statutory provisions? | EH&S policy was as per guidelines. | --- |
| 4. | When was the OH & S policy declared and adopted? | EH & S policy declared and adopted in 2014. | --- |
| 5. | Whether the OH & S policy reviewed periodically? | EHS policy reviewed periodically. | --- |
| 6. | Whether the OH & S policy is available in local language and made known to all? | EHS policy was available in English language only. | 82. As per rule 68-O of the GFR-1963, OH&S policy to be made available in a language understood by the majority of workers and explained to them. |
| 7. | What was the last date of updation? | Last updation was on dtd. 01/04/2022. | --- |
| | **C-2 OH & S Organizational Set Up** | | |
| | **C-2.1 Safety Department - The FA Section 40-B** | | |
| 8. | Does the factory have a safety department and what is strength of safety department? | Factory has a safety department and strength of Safety department was 9 persons (Including safety offices, EHS Manager, EHS Head, EHS Engineer & Firemen). | --- |
| 9. | Whether the strength and qualifications of Safety Officers are as per the statutes? | We observed that, strength and qualifications of Safety Officers were as per the statutes Notification No. **KHR-529-FAC-1077-36785-M-(1)** dated 19.12.1985 | --- |
| 10. | Does the head of safety department report to the Chief Executive? | The Safety officer report to EHS manager, EHS head & Unit head. | --- |
| 11. | How often are the safety officers retrained in the | Safety officer trained in periodic refresher courses | --- |

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| | latest techniques of total safety management? What is the frequency of retraining? | & participated in various activities taken up by the Factory inspectorate and local agencies | |
| 12. | What additional duties the safety officer is required to do? | Only Safety related Functions. | --- |
| 13. | What is the power of safety officer vis-a-vis unsafe condition or unsafe act? | The power of safety officer is to stop any dangerous operation. If there are any unsafe conditions or acts, these will be brought to the notice of the concerned HOD and in the case of imminent danger, the job stopped until unsafe conditions / act was removed / corrected. | --- |
| | **C-2.2 Safety Committee(s) (The FA Section 41-G, Rule 68-F, Y of the GFR 1963)** | | |
| 14. | Does the factory has a safety committee(s)? What are the types, structures and terms of reference of the committees? | Safety Committee meeting held quarterly basis.<br>Last Safety Committee Meeting held on – 18/01/2024<br>Total – 28 Members<br>Chairman – Mr. D. N. Rai<br>Secretary – Mr. Ashish Joshi | --- |
| 15. | Is the constitution of the safety committee(s) as per the statute? | Constitution of the safety committee(s) was as per the statute. | --- |
| 16. | How are the members of safety committee(s) selected? (elected / nominated) | Nominated | --- |
| 17. | How often are the meetings of safety committee(s) held? | Meetings of safety committee(s) held once in a quarter. | --- |
| 18. | Are the recommendations of the committees(s) implemented? | The recommendations of the committees(s) were implemented by action plan by concern person in time bound manner. | --- |
| 19. | Are the minutes of the safety committee(s) meetings circulated among the members? | MOM of safety committee was circulated to members through mail & displayed on notice board. | --- |
| 20. | Are the minutes forwarded | MOM of safety committee | --- |

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| | to the trade union(s) and chief executive and occupier? | was circulated to committee members and Unit head. | |
| 21. | Whether the management and trade union play their active roles in supporting and accepting the committee(s) recommendations? | Its informed that management support in compliance of recommendations of safety committee. | --- |
| 22. | How are the safety committee(s) members apprised of the latest development in safety, health and environment? | Members of the safety committee's members were participating in safety training. | --- |
| | **C-2.3 Safety Budget (The GFR- Rule 12-C & 68-0)** | | |
| 23. | What is the annual safety budget? | EHS Budget was available | --- |
| 24. | How much percentage is this budget of the total turnover of the company? | Adequate - made available as and when required. | --- |
| 25. | How much budget has been utilized till date? | Data available at EHS department. | --- |
| 26. | Is the safety budget adequate? | It's informed that Safety budget was adequate. | --- |
| 27. | How is the safety budget arrived at? | Based on the previous year's utilization & present requirements. | --- |
| 28. | What is the pattern of expenditure for the last five years? | Mostly Increasing trend. Price hike & the requirement. | --- |
| 29. | What are the approved sanctions for the expenditure in this budget? | Data available with EHS department | --- |
| 30. | Does this budget get reflected in the annual report of the company? | We came to know that budget get reflected in the annual report of the company. | --- |
| | **C-3 Safety Manual (The FA Section 41-B,41-C,41-H, GFR-Rule 68-K)** | | |
| 31. | What is the periodicity of updation / review of safety manual? | Safety booklet was not evident. | 83. Safety booklet to be prepared and made known to all. |
| 32. | Does the safety manual adequately address all the hazards in the plant? | Safety booklet was not evident. | --- |
| 33. | Are the employee made aware of safety rules? | Safety booklet was not evident. | --- |
| | **C-4 Standard Operating Procedures (SOP)** | | |

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|---|---|---|---|
| 34. | Are written Standard / safe operating procedures available for all operations and processes? | Standard operating procedures were available for operations and processes. | --- |
| 35. | Whether the written Standard / safe operating procedures are displayed or made available and explained in the local language to the workers? | The written Standard / safe operating procedures were displayed or made available in the english only. | 84. Standard / safe operating procedures should be made available in local or national language and displayed it at conspicuous places. |
| 36. | Whether concerned section and safety department prepares standard / safe operating procedure jointly? | Concerned section and safety department prepares standard / safe operating procedure jointly. | --- |
| 37. | Are standard / safe operating procedures reviewed and updated? | Standard / safe operating procedures reviewed periodically and updated by concerned section. | --- |
| 38. | Have the workers been informed of the consequences of failure to observe the standard /safe operating procedures? | Workers were informed about the consequences of failure to observe the standard / safe operating procedures during training & TBT. | --- |
| | **C-5 Plant Modification Procedures** | | |
| 39. | What is the system for effecting any change in the existing plant, equipment or process? | MOC (Management of Change) system for controlling any change in the existing plant, equipment or process was not evident. | 85. MOC / Change control procedure to be prepared for any change in plant, equipment or process. |
| 40. | Whether the P & I diagram, and other related documents are updated accordingly? | Its inform that, P & I Diagram and other related documents were updated. | --- |
| 41. | Whether hazard assessment done before implementation of modification? | MOC (Management of Change) system was not evident. | --- |
| | **C-6 Work Permit System (The GFR- Rule 102 Chemical Works Sch. 19)** | | |
| 42. | What types of work permits exist in the factory? | Record of work permits were available.<br>**List of work permits:**<br>• Hot work<br>• Confined Space | --- |

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|---|---|---|---|
| | | • LOTO<br>• Cold work<br>• Height Work<br>• Excavation<br>• Line opening<br>• Hazardous chemical loading / unloading | |
| 43. | Are the necessary forms detailing required safety precautions have been prepared and used for each type of work-permit? | Necessary forms detailing required safety precautions have been prepared and used for each type of work-permit.<br>Printed Format in Original, Duplicate & Triplicate was developed for all permit | --- |
| 44. | Is the responsibility assigned to authorized person for issuing of safety work permit? | Responsibility assigned to authorized person for issuing of safety work permit | --- |
| 45. | Is the copy of safe work permit sent to safety officer before execution of the job? | One copy of work Permit sent to safety Officer before execution of the job. | --- |
| 46. | Is validity period specified in the safety work permit? | Date, time and location, validity was mentioned in the safety work permit. | --- |
| 47. | Are the records of work permit available and maintained in proper order? | Records of work permit was available and maintained in proper order<br>Record was available at EHS department. . | --- |
| | **C- 6.1 Control Measures for Work at Height** | | |
| 48. | Is adequate safe access provided to all places where workers need to work? | Adequate access provided where workers need to work. | --- |
| 49. | Are all such access in good condition? | Most of access was found in good condition. | --- |
| 50. | Are all scaffolds are properly designed and erected? | Evident of scaffolding uses was not found.<br>As & when required, erected through contractor only. | 86. When scaffolding installed, its guideline should be followed |
| 51. | Are scaffolds inspected every day before work begins? | Evident of scaffolding uses was not found. | --- |
| 52. | Are ladders securely | Evident of scaffolding uses | --- |

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| | clamped or lashed in place? | was not found. | |
| 53. | Are planks in good condition? | Evident of scaffolding uses was not found. | --- |
| 54. | Are scaffold walkaways, platforms, runs or stairs free of debris, grease, any unnecessary obstruction and projecting nails? | Evident of scaffolding uses was not found. | --- |
| 55. | Are the scaffolds higher than 20 m.? If yes, is netting or intermediate railing provided between toe-boards and hand railings? | No scaffolds higher than 20 m.<br>Third party hired for above 20 m height work. | --- |
| 56. | Are folding stepladders properly used? | Folding stepladders used properly. | --- |
| 57. | Are ladders set up at the proper slope of about 1:4? | Ladders set up proper slope of about 1:4 Ratio. | --- |
| 58. | Do workers use hand lines to lift tools or materials? | Workers used hand lines to lift tools & materials. | --- |
| 59. | Are proper ladders used around electrical hazards? | Proper ladders used around electrical hazards was not evident. | 87. Special ladders should be used around electrical hazards. |
| 60. | On sloping roofs, are crawling boards, lifelines, safety belts and edge protection provided where needed? | Safety belt and edge protection was provided where necessary. | --- |
| 61. | Whether the weak spots, skylights, or deteriorated asbestos-cement boards through which a worker might fall while working in the roof has been identified and safety net provided appropriately? | Week spot, skylights, or deteriorated asbestos-cement boards was not seen. | --- |
| 62. | Are the workers being medically examined for their fitness to work at height? | The workers medically examined for their fitness to work at height was not evident. | 88. Medical examination records to be made available for fitness to work at height. |
| | **C-6.2 Work in Confined Space (The FA Section 36, GFR-Rule 102 Schedule 19)** | | |
| 63. | Is work permit system followed for working in confined space? | Confined space permit system followed for working in confined space. | 89. As per Rule 102, Schedule 19, Part II, Para (18) (1) (a) of The Gujarat Factories |

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| | | | Rules, 1963, Confined pace entry to be identified and to be displayed.<br>90. Confined space entry to be done with competent person authorization only.<br>91. Kindly refer Chapter-8 "Annexure-A Area of Improvements" |
| 64. | Whether monitoring of the atmosphere inside the confined space is carried out and ensured that there is no flammable or toxic gas in the area? | Monitoring of the atmosphere inside the confined space was carried out. ($O_2$, LEL, etc. checked) | 92. Confined space entry to be monitored by DISH approved competent person only. |
| 65. | Whether the person entering the confined space is using suitable personal protective equipment (PPE)? | The person entering the confined space used all required personal protective equipment. (Helmet, SCBA set, gloves, goggles, safety shoes, etc.). | --- |
| 66. | Is rescue team available in case of any emergency? | Rescue team available immediately in case of any emergency.<br>Rescue team was available round the clock in organization. | --- |
| | **C-7 Contractors' Safety System** | | |
| 67. | Is there any system for selection of contractors? | Contractors were selected upon fulfilling the pre-qualification criteria. Contractor management system was available with HR department. | --- |
| 68. | Are there any guidelines on contractor's safety and training? | Guidelines on contractor's safety and training record was not evident. | 93. Guidelines on contractor's safety and training to be made. |
| 69. | Whether contract document includes necessary safety and welfare clauses as per statutes? | Yes, necessary documents were available. | --- |
| 70. | Is there any programme to ensure use of PPE by | Training, supervision and site monitoring to ensure | --- |

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|---|---|---|---|
| | contractor's personnel? | use of PPE by contractors' personnel. | |
| 71. | Do the contractors have their own safety organization? | They were trained & experienced. They co-ordinate with factory Safety Dept. | --- |
| 72. | Are the contractors reporting all accidents and injuries? | The contractors reporting accidents and injuries. | --- |
| 73. | Are contractor workers trained to observe safety at workplace? | Contractor workers trained by safety officer to observe safety at workplace. | --- |
| 74. | Whether contractor workers are engaged in process / operations? If yes, are they aware of safe operating procedures? | Contractor workers were engaged in process / operations.<br>Contractor workers were aware of safe operating procedures by training. | --- |
| | **C-8 Plant Design and Layout (The FA Section 87, GFR-Rule 102)** | | |
| 75. | Whether hazardous operations in the plant are segregated? | Hazardous operations in the plant were segregated. | --- |
| 76. | Whether occupational health & safety aspects are considered during the design? | Occupational health & safety aspects were considered during the design | --- |
| 77. | Are all the equipment provided with adequate space for working, maintenance etc.? | The equipment provided with adequate space for working, maintenance, etc. | --- |
| 78. | Are the storage tanks provided with enough space Clearance between them? | The storage tanks provided with enough space clearance between them. | --- |
| 79. | Whether the plant layout has taken care of the movement of firefighting equipment and emergency exits? | The plant layout has taken care of the movement of firefighting equipment and emergency exits. | --- |
| | **C-9 Medical Management of Accidents (The FA Section 45, GFR-Rule 68-U,68-V,71)** | | |
| 80. | Are medical facilities available with trained first aid staff and equipment in round the clock shift for all including contractors? | Occupational health centre: available<br>FMO - Dr. Tejas Mehta<br>First aider - 10 Nos. | 94. OHC to be reviewed as per The Rule – 68 U of The Gujarat Factories Rule 1963. |

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| 81. | Is the ambulance van available for round the clock basis with the dedicated driver? | Dedicated vehicle was available round the clock basis with dedicated driver. | --- |
| 82. | Is there any mutual aid scheme available with the nearest hospitals to manage and treat injuries during emergency? | Yes, mutual aid scheme available with nearby Hospital – Jolva Hospital, Dahej. | --- |
| 83. | Are the workers / contractor workers aware of emergency medical facilities? | Workers / contractor worker aware about the emergency medical facilities. | --- |
| | **C-10 Management of Emergencies (Natural / Man-Made)** | | |
| 84. | Does the system exist to detect and control these Emergencies? | Yes, Emergency Response Plan (OSEP) plan was in place.<br>Fire protection system, Fire alarm system & panel control, Mock drill & OSEP available for detect and control emergencies. | --- |
| 85. | Are the employees aware of the measures to be taken during emergencies? | Workers found aware about the measure to be taken during emergencies.<br>Regular basis training given & mock drill carried out. | --- |
| | **C-11 Employees Selection and Placement** | | |
| 86. | Whether norms are available for selection of different category of employees? | Job Description criteria available with HR Department for selection of different category of employees. | --- |
| 87. | Whether pre-employment medical examination is being conducted for employees? | Pre-employment medical examination (Form No. 33) Maintained. | --- |
| 88. | Is there any procedure to evaluate safety awareness and record of the employees during their promotion? | Performance managment system available for promotion. | --- |
| | **C-12 Safety Culture** | | |
| | **C-12.1 Attitudes of Managers** | | |
| 89. | Do the managers follow the plant safety rules at all | Managers follow the plant safety rules. | --- |

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| | times? | | |
| 90. | What are their attitudes towards safety reviews and audits? | Positive attitudes towards safety reviews and audits. | --- |
| 91. | What is the response of management to safety violation? | Safety Violation system available. | --- |
| 92. | Whether safety related decisions are taken in consultation with the workers? | Safety related decisions were taken in consultation with the workers. | --- |
| 93. | What is the attitude of the managers towards non-use of personal protective equipment? | Manager Acts as per Safety violation policy & norms, non-use of personal protective equipment. | --- |
| | **C-12.2 Attitudes of Workers** | | |
| 94. | Whether workers are aware of the consequences of their wrong actions? | Workers were informed during Tool Box Talk (TBT) / training / plant round about the consequences of their wrong actions. | --- |
| 95. | Are laid down safe working procedures followed strictly? | Safe working procedures followed. | --- |
| 96. | What is the attitude of the workers towards their own mistake, which can prejudice safety? | We came to know that workers follow safety rules and procedures. If deviation observed by anybody they will accept and correct it. | --- |
| 97. | Do the workers report near miss incidents and suggest safety improvements? | Near miss incidents reporting system was available. | --- |
| 98. | Are the workers aware of the system of rewards and sanctions relating to safety matters? | Workers were adequately aware about the safety rewards. | --- |
| 99. | What is the attitude of workers towards use of personal protective equipment? | Positive attitude of workers towards uses of personal protective equipment. | --- |
| | **C-13 Statutory Licenses, Approvals and Records** | | |
| 100. | Whether all the safety related Acts / Rules (with latest amendments) applicable to your organization identified, | Legal register with all applicable Acts /Rules identified, maintained, amended time to time and communicated with all | --- |

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| | informed to all employees and complied? | concerned.<br>Kindly refer Chapter-7 "Applicable Laws and Statutory Requirements" & Chapter-9, "Annexure-B: Examination of Records (IS 14489:2018)" | |
| 101. | Whether the licenses have been validated? | All licenses found in valid status.<br>For Details kindly refer "Annexure-1 List of Licenses/ Certification and its validity ". | --- |
| | **C-14 Motivational and Promotional Measures for OH & S** | | |
| 102. | Does the factory have occupational health and safety suggestion scheme? | Factory have occupational health and safety suggestion scheme. | --- |
| 103. | Are occupational health and safety contests organized in the factory? | Yes, During safety week celebration | --- |
| 104. | Does the factory participate in National Awards? | No | --- |
| 105. | Has the factory been awarded during last five years? | No | --- |
| 106. | Does the organization publish safety bulletin /Newsletters? | Organization not published safety bulletin /Newsletters but some safety related information was circulated through mail. | --- |
| 107. | Whether the safety bulletins are widely distributed? | Yes, safety information was widely distributed through mail & displayed on notice board. | --- |
| 108. | How is the occupational health and safety information including accident statistics disseminated in the factory? (Bulletin boards, Newsletter etc.) | The occupational health and safety information including accident statistics disseminated in the factory by Bulletin boards. | --- |
| 109. | What are the activities conducted during National Safety Day / week? | Safety quiz, firefighting competition, first aid training, PPE exhibition, SCBA set competition, housekeeping competition, safety slogan, safety poster, poem, etc. | --- |

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| 110. | What is the percentage of Workers participating in the various safety promotional activities? | Majority of Workers participating in the various safety promotional activities. | --- |
| | **C-15 Hazard Identification and Job Safety Analysis** | | |
| 111. | Was an initial process hazard analysis (PHA) completed? | Yes, HAZOP, QRA, etc. carried out. | 95. Please ensure that all recommendation of PHA to be complied with.<br>96. Ensure that PHA to be updated after any change in the process. |
| 112. | What are the stages of PHA? Whether a dedicated group is identified for PHA? | Two Stages of PHA.<br>PHA carried out by dedicated team. | --- |
| 113. | Was the PHA appropriate for the complexity of the process and identify, evaluate, and control the hazards involved in the process? | The PHA (HAZOP) study was found appropriate for the complexity of the process and identify, evaluate, and control the hazards involved in the process. | --- |
| 114. | Does the hazard evaluation use one or more of the following PHA methodologies: What-If Analysis, Process Checklist, Hazard and Operability Study (HAZOP), Failure Mode and Effects Criticality Analysis (FMECA), Fault Tree Analysis (FTA) or any other appropriate equivalent methodology? | HAZOP study of process plant was carried out. | --- |
| 115. | Does PHA assure addressing issues of inherent safety features with respect to material and their properties? | Yes, covering inherent safety features with respect to material and their properties. | --- |
| 116. | Does the PHA address the hazard identification, incidents history, consequences of failures (engineering and administrative controls), human factors, consequent analysis with respect to | PHA addressed the hazard identification, incidents history, consequences of failures (engineering and administrative controls), human factors, consequent analysis with respect to | --- |

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| | possible safety and health effects of failure of controls? | possible safety and health effects of failure of controls. | |
| 117. | What are the stages of PHA? Whether a dedicated group is identified for PHA? | Two Stages of PHA.<br>PHA carried out by dedicated team. | --- |
| 118. | Does the system exist to promptly address findings and recommendations of PHA? | Yes, system exist to promptly address finding & recommendations of PHA. | 97. Please ensure that all recommendation of PHA to be complied with. |
| 119. | Are the PHA's updated and revalidated at least every five years by a qualified team to assure that the PHA is consistent with the current process? | PHA updated & revalidated by a qualified team to assure that the PHA is consistent with the current process. | 98. Cyclic HAZOP to be done on regular basis and records to be done. |
| 120. | Whether the activities requiring Job Safety Analysis have been identified? | Activities identified for Job Safety Analysis. | --- |
| 121. | Whether the identified jobs for Hazard Identification have been carried out by trained and experienced persons? | The identified jobs for Hazard Identification was carried out by internally trained and experienced persons. | --- |
| 122. | Whether the checklists have been prepared on each Job Safety Analysis and are being used while carrying out the job? | The checklists was not prepared for Job Safety Analysis. | 99. Checklist to be prepared for the Job Safety Analysis. |
| | **C-16 Product Safety<br>(Dangerous Goods {Classification, Packaging and Labelling} Rule, 2013)** | | |
| 123. | Whether hazards arising from use of the products are identified? | Hazards arising from use of the products were identified. | --- |
| 124. | Whether material safety data sheet prepared for the products? | Material safety data sheet prepared for the products. | --- |
| 125. | Are all the products labelled and packed appropriately? | All the products labelled and packed appropriately. | --- |
| 126. | Whether safety instructions are given along with products? | Mandatory safety instructions including TREM Card / MSDS etc. are given along with products. | --- |
| 127. | Whether policy exists for recall of products? | Process was available for recall of products. | --- |

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| | **C-17 Safety Training (The FA 1948-Sec 111(A), 7 (A))** | | |
| 128. | Whether training needs have been identified? | Training needs identification based on the skill matrix and accordingly training provided. | --- |
| 129. | Is there any programme of induction training, its duration and topics covered? | Induction training was provided to all newcomers. | --- |
| 130. | Whether the assessment of the trainees has been carried out? | The assessment of the trainees has been carried out by verbally. | --- |
| 131. | What are the infra-structural facilities available for training? | Conference room, Projector, Table, Chair, etc. available. | --- |
| 132. | Whether training is conducted by qualified person? | Training was provided by Safety Officer internally & also from outside agency. | --- |
| 133. | Whether trainers are being re-trained from time to time? | Trainers were being re-trained from time to time by internally and outside agency. | --- |
| 134. | Whether proper records of training program conducted are maintained? | Training records of training program conducted were maintained | --- |
| 135. | How training programs are evaluated? | Training programs were evaluated through traning feedback. | --- |
| 136. | Whether schedule for training on occupational health and safety is available and maintained? | Yearly training Calendar prepared and accordingly trainings were conducted. | --- |
| 137. | Whether the training programmes are reviewed? | Regular basis training programmes were reviewed. | --- |
| 138. | Are all the employees periodically trained / retrained and what is the frequency of such training? | As per monthly training schedule all the employees trained / retrained. | --- |
| 139. | Are the retraining needs identified whenever a new process / products and change in existing process introduced? | Retraining needs identified whenever a new process / products and change in existing process was introduced. | --- |
| 140. | Whether training covers top management? | Training covers employees of factory and top management too. | --- |

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|---|---|---|---|
| 141. | How many hours of safety training is given to different employees? | Data available with EHS / HR department. | --- |
| | **C-18 Change Management** | | |
| | **C-18.1 Management of Change** | | |
| 142. | Are there written procedures for managing change to process chemicals, technology, equipment and procedures and changes to facilities that affect the plant process / system operation? | Informal procedure for management of change. | 100. MOC (Management of Change) Procedure to be made available. |
| 143. | Do the procedures assure that the technical basis for the proposed change addressed prior to any change? | Informal system for management of change. | 101. MOC (Management of Change) Procedure to be reviewed, implemented and record to be maintained for all changes. |
| 144. | Do the procedures assure that the impact of the change on safety and health addressed prior to any change? | Informal system for management of change. | --- |
| 145. | Do the procedures assure that modifications to operating procedures are addressed prior to any change? | Informal system for management of change. | --- |
| 146. | Do the procedures assure that the necessary time period for the change is addressed prior to any change? | Informal system for management of change. | --- |
| 147. | Do the procedures assure that the authorization requirements for the proposed change are addressed prior to any change? | Informal system for management of change. | --- |
| 148. | Are employees involved in operating a process, and maintenance and contract employees whose job tasks will be affected by change informed of, and trained in, the change prior to the | Informal system for management of change. | --- |

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| | start-up of process or affected part of process / operations? | | |
| 149. | Is the safety information is reviewed and updated on changes? | Informal system for management of change. | --- |
| 150. | Are the operating procedures or practices updated? | Yes, Operating procedures / practices updated. | --- |
| | **C-18.2 Mechanical Integrity** | | |
| 151. | Does the mechanical integrity program include for all mechanical equipment including pressure vessels and storage tanks, piping and components, relief devices and vent systems, emergency shutdown systems, pumps, control systems? | Mechanical equipment including pressure vessels and storage tanks, piping and components, relief devices and vent systems, emergency shutdown systems, pumps, control systems included in mechanical integrity program.<br>Testing done as per norms.<br>Form No. 9, 10 & 11 and Stability of the Building available and valid. | 102. Equipment / instrument inspection to be regularly done. |
| 152. | Are there written procedures to maintain the on-going integrity of process equipment? | Its inform that written procedure was available to maintain the on-going integrity of process equipment.<br>Preventive maintenance schedule was available. | --- |
| 153. | Whether training been provided to each employee involved in maintaining the on-going integrity of process equipment? | Training was imparted to employees on PM of equipment. | --- |
| 154. | Are inspections and tests performed on each item of process equipment included in the program? | Examination, inspection and testing carried out of process equipment. | 103. All pipelines carrying flammable or explosive substances to be examined by responsible person once in week to detect any deterioration or defect. |
| 155. | Does the inspection and test frequencies meet the manufacturer's recommendation and good | Inspection and test frequencies meet the manufacturer's recommendation and good | --- |

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|---|---|---|---|
| | engineering practice? | engineering practice. | |
| 156. | Are inspections and tests performed more frequently if determined necessary by operating experience? | Inspections and tests performed more frequently when determined necessary by operating experience. | --- |
| 157. | Are deficiencies in equipment that are outside limits corrected before further use so as to assure safe operation? | Deficiencies in equipment that are outside limits corrected before further use so as to assure safe operation. | --- |
| 158. | In the construction of new plants and equipment, whether quality assurance programme is implemented to ensure that equipment fabricated is suitable for the process? | Quality assurance programme was available for construction of new plant, equipment. | --- |
| 159. | Are appropriate checks and inspections made during equipment installation stage? | Appropriate checks and inspections made during equipment installation stage. Information available with engineering department. | --- |
| 160. | Are the suitability of maintenance materials, spare parts and equipment ensured during maintenance? | Yes, suitability of maintenance materials, spare parts and equipment ensured during maintenance. | --- |
| | **C-19 Physical Hazard** | | |
| | **C-19.1 Housekeeping (The GFR – Rule 16, 17, 17 A, The FA 1948 section 11.)** | | |
| 161. | Are all the passages, floors and the stairways in good condition? | The passages, floors and the stairways in good condition at most of place. | --- |
| 162. | Is glass door taped or otherwise marked to make it visible to workers? | Glass door was taped / marked. | --- |
| 163. | Do you have the system to deal with the spillage? | Soda Ash, sand bucket, lime, spill kit, etc. available to deal with the spillage. | --- |
| 164. | Do you have sufficient disposable bins clearly marked and whether these are suitably located? Are containers of refuse (waste) and trash emptied at the end of every day or | Sufficient disposable bins clearly marked and suitably located. All containers or bins regularly cleaned. | --- |

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| | soon after they are full? Are the containers or bins regularly cleaned? | | |
| 165. | Are drip trays positioned wherever necessary? | Drip trays were positioned wherever necessary. | --- |
| 166. | Do you have adequate localized extraction and scrubbing facilities for dust, fumes and gases? Please specify. | Fume Extract System, Scrubber installed. Detailed information available with HSE and concern plant. | --- |
| 167. | Whether walkways are clearly marked and free from obstruction? | Walkways were marked. | 104. It is recommended that walkway to be clearly marked and free from obstruction all the time. |
| 168. | Do you have any inter-departmental competition for good housekeeping? | We came to know that Inter-departmental competition was carried out for good housekeeping. | --- |
| 169. | Has your organization elaborated good housekeeping practices and standards and made them known to the employees? | Housekeeping needs improvement. | 105. Good housekeeping practices and standards should be documented and to be made known to the employees. |
| 170. | Are there any working conditions, which make the floors slippery? If so, what measures are taken to make them safe? | Working conditions, which make the floors slippery was available and adequate measures taken. | 106. Efficient floor cleaning programs to be introduced. |
| 171. | Does the company have adequate measures to suppress polluting dust arising out of materials stored on the roadside? | Some unwanted material found on road. | 107. Internal roads to be kept clean & clear. Unwanted material to be removed from internal road. |
| | **C-19.2 Machine and General Area Guarding (The GFR – Rule 54, The FA 1948 Section – 21, 22.)** | | |
| 172. | Whether machinery and equipment which can cause physical injuries to operator have been identified? | Machinery & equipment which can cause physical injuries to operator was not identified. | 108. Machinery & equipment which can cause physical injuries to operator to be identified. |
| 173. | Are all moving parts and point of operation of machinery adequately guarded? | We observed that at some locations guard was missing. | 109. All moving parts to be adequately guarded. |
| 174. | Are all fixed guards securely bolted in position | Fixed guards were securely bolted in position and in | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | and in good condition? | good condition. | |
| 175. | Are all interlock guards for prevention of physical injury in good condition? | Interlock guards provided for prevention of physical injury and observed in good condition. | --- |
| 176. | Are all emergency stop buttons effective and clearly labelled? | Emergencies stop buttons clearly labelled. | 110. Ensure that all emergency stop buttons to be labelled properly. |
| 177. | Are the operators for machines having moving parts aware of the danger of working with loose clothing? | Operators for machines having moving parts aware by training of the danger of working with loose clothing. | --- |
| 178. | Are the openings where there is free fall hazard covered or fenced securely? | No openings available where there was free fall hazard. | --- |
| | **C-19.3 Material Handling (The FA 1948 section 28,29,34, The GFR – Rule 58,59,60,62)** | | |
| 179. | Are adequate equipment available for handling materials? | Goods lift, Forklift, Stacker, hand trolley, etc. available for handling materials. Manual material handling also carried out. | 111. Ensure that material handling equipment to be regularly inspected and maintained. |
| 180. | Are the workers aware of the hazards associated with material being handled? | Workers found aware by training of the hazards associated with material being handled. | --- |
| 181. | Where manual handling is necessary, are the workers been trained? Do they practice this? Are workers warned for lifting of excessive weight? (Maximum weight of material for adult male and female are 55 Kg and 30 Kg respectively) | Workers trained where manual handling necessary.<br>All workers warned for lifting of excessive weight. | --- |
| 182. | Do workers follow safe procedures for storage of materials? | Workers followed safe procedures for storage of materials. | --- |
| 183. | Is the register maintained to record particulars of examination of all lifting machines, tools and tackles? | Register was maintained to record particulars of examination of all lifting machines, tools and tackles.<br>(Ref.: Form No. 9, 10 | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | | available and valid) | |
| 184. | Are all the statutory examinations and tests carried out and certified by competent person(s)? | All the statutory examinations and tests were carried out and certified by competent person(s) by DISH Gujarat. | --- |
| 185. | Are the operators of crane, lifts, hoists and other mechanized operations adequately qualified? | We observed that operators of mechanized operations adequately qualified.<br>Record available at concern department. | --- |
| 186. | Is the safe working load clearly marked? | SWL was marked on most of the equipment | 112. To be ensured Safe working load (SWL) clearly marked. |
| 187. | Has the person employed to operate crane, forklift, or to give signals to crane been medically examined for eyesight and colour vision? | During periodic medical examination eye-sight and colour blindness was checked. | --- |
| 188. | Is the frequency of eyesight and colour vision examination as per the latest rules? | Frequency of eyesight and colour vision examination as per the latest rules. | --- |
| | **C-19.4 Electrical Safeguarding** | | |
| 189. | Are licensed electricians available for electrical work? | We came to know that Licensed electricians were available for electrical work. | --- |
| 190. | Whether area classification for electrical equipment has been carried out? | Area classification for electrical equipment was carried out and accordingly electrical fittings were provided. | --- |
| 191. | Do the electrical fittings conform to area classification for electrical equipment? | Flame-proof electrical installations were installed in plant, storage tanks, warehouses, etc. | 113. Ensure electrical installations are confirming to hazardous area classification. |
| 192. | Is a ground fault current interrupter system (ELCB) in use? | ELCB in use. | --- |
| 193. | Are all connections made by using appropriate plugs, receptacles or enclosures? Are fuses provided? | All connections made by using appropriate plugs. | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| 194. | Are there any makeshift connection bare wires or damaged cables? | Shift connection bare wires or damaged cables was not observed. | --- |
| 195. | Is there a system of ensuring periodical inspection of hand tools, extension boards used for electrical work? | Periodical inspection of hand tools, extension boards, etc. carried out and checklist / records were maintained. | --- |
| 196. | Do the workers use proper types of PPE during the working on live line? | Workers used proper types of PPE during the working on live line. | 114. Ensure that Electrical Safety Shoes, Safety Helmet, Safety Goggles, Arc Flash Suit of suitable rating 11KV hand glove and Shoes, Hot line stick tester used. |
| 197. | Is the separate work permit issued for working on high voltage line? | No separate work permit issued for working on high voltage line.<br>LOTO available and permit system for temporary connection. | 115. Separate Permit to be issued for high voltage Electrical line.<br>116. Review CEA regulation. |
| 198. | Whether the process(s) and equipment that generate and accumulate static charge have been identified? | Most of Processes, operations and equipment generating static charge were identified. | 117. Process(s) and equipment that generate and accumulate static charge to be identified |
| 199. | Whether all such equipment including pipelines for flammable materials are properly bonded and earthed? | Such equipment including pipelines for flammable materials were properly bonded and earthed. | --- |
| 200. | Whether earth pit resistance is measured, and the record maintained? | Earth pit resistance was measured and the record was maintained by electrical department. | --- |
| 201. | Whether lightning arrestor has been installed and is adequate? | Lightning arrestors installed. | 118. Lightning arrestor survey to be carried out. |
| | **C-19.5 Safety in Storage and Warehousing** | | |
| 202. | Whether the Material Safety Data Sheet for all chemicals is available? | Material Safety Data Sheet for chemicals was available. | --- |
| 203. | Are the chemicals stored as per their hazardous properties including the incompatibility? | Most of the Chemicals stored as per their hazardous properties including the incompatibility. | 119. Chemical compatibility chart to be displayed |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| 204. | Are all containers clearly, indelibly labelled? Are all chemicals stored as per safety regulations? | The containers clearly, indelibly labelled & Identification and hazard symbol was available on container. | --- |
| 205. | Whether all racks and steel cages have sufficient load bearing capacity? | The racks and steel cages have sufficient load bearing capacity. | 120. Ensure that SWL to be marked on racks. |
| 206. | Is adequate natural ventilation provided to storeroom? Is there any emergency exit? | Adequate natural ventilation provided to storeroom.<br>Emergency exit provided. | --- |
| 207. | Whether adequate firefighting arrangement existing in flammable chemical storage? | Fire Hydrant, Fire Extinguisher, Sprinkler, sand buckets, etc. available. | --- |
| 208. | Whether methodology for handling spillages of hazardous chemical available along with the equipment required handling the spillage? | Procedure was available for handling spillages of hazardous chemical available along with the equipment required handling the spillage. | --- |
| 209. | Whether aisles are marked, and emergency exits displayed? | Aisles were marked and emergency exits were displayed. | --- |
| | **C-19.6 Hazard Assessment for New Equipment** | | |
| 210. | What is the system for effecting any change in the existing plant, equipment? | Informal system available for effecting any change in the existing plant, equipment was available. | 121. MOC / Change Control Process to be established. |
| 211. | Is there system for evaluating hazards from new equipment? | System available for evaluating hazards from new equipment. | --- |
| 212. | Whether the P and I diagrams, and other related documents are updated accordingly? | We came to know that P and I diagrams, and other related documents were updated accordingly. | --- |
| 213. | Is any Job Hazard Analysis (JHA) carried out after installation of new equipment? | Informal JHA carried out for non-routine activity. | 122. JHA to be carried out after installation of new equipment. |
| | **C-19.7 Hazards from Radiation Sources** | | |
| 214. | Whether licenses have been obtained for storage /handling of radioactive material? | We came to know that storage of radiation sources was not available.<br>As & when required (For NDT testing or other work) | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | | handling of radioactive material used by third party, they had valid license for handling of radioactive material. | |
| 215. | Whether approved Radiological Safety Officer appointed? | As & when required (For NDT testing or other work) approved Radiological Safety Officer appointed. | --- |
| 216. | Whether appropriate PPEs are used against radiation hazards? | As & when required (For NDT testing or other work) Radiological Safety Officer used appropriate PPEs against radiation hazards. | --- |
| 217. | Is the flooring of the radioactive material handling area amenable for proper decontamination? | Flooring of the radioactive material handling area was proper decontamination by third party which carried out job. | --- |
| 218. | Is the storage room of radiation source as per the licence condition? | They don't have storage of Radioactive material in the premises. | --- |
| 219. | Are all persons working in the facility have radiation safety training? | We came to know that as & when required NDT testing or other work only trained persons & qualified person handled working. | --- |
| 220. | Is the operator handling devices using radioactive materials qualified and possess the necessary certificate? | As & when required (For NDT testing or other work) only trained persons & qualified person handling of radioactive materials. | --- |
| 221. | Is the periodical radiation monitoring carried out? | As & when required (For NDT testing or other work) radiation monitoring carried out | --- |
| 222. | Are the records of inventory of radioactive material maintained in the standard format and submitted to the competent authority as per the period specified? | We came to know that third party (which carried out job) was do all records of inventory of radioactive material maintained in the standard format and submitted to the competent authority as per the period specified | --- |
| 223. | Are emergency handling tools available? | We came to know that third party (which carried out job) had emergency | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | | handling tools available. | |
| 224. | Are the personnel monitoring badges (TLD, Pocket dosimeter etc.) assigned and worn by each radiation worker? | We came to know that third party (which carried out job) worn suitable personnel monitoring badges. | --- |
| 225. | Are the radiation symbol and red light displayed as required? | We came to know that, radiation testing done in lunchtime & all area cordon by barricaded line. | --- |
| | **C-20 Chemical Hazard** | | |
| | **C-20.1 Transportation of Hazardous Substances** | | |
| 226. | What potentially hazardous materials are transported to or from the site (including wastes) | List of Hazardous material. Kindly see para 6.2 "List of Hazardous Raw Materials". (Isobutylene, n-Hexane, Toluene, Acrylonitrile, Triethyl Amine, Ethanol, IPA, EDC, Methanol, Benzaldehyde, DMF, MEG, Phenol, Carbon Tetrachloride, Bromine, Chlorine, Caustic Lye, HCl, Sulphuric Acid, Spent Acid, Formic Acid, TC, NSS, C-9, O-Xylene, MDC, Benzene, Ethyl Acetate, Acetic Acid, Mandelo Nitrile, Hydrazine Hydride, Aluminium Chloride, Sodium Cyanide, Acetonitrile, DMA, ODCB, 2,5 DCA, $SO_2$, $CO_2$, Methyl Chloride, HSD) | --- |
| 227. | What mode of transport are used?<br>1) Road,<br>2) Rail, and<br>3) Pipelines | Mode of transport was used<br>• Road<br>• Pipeline | --- |
| | **1) Road** | | |
| 228. | Does the company employ licensed vehicle of its own / outside sources? | Outside sources vehicle. | --- |
| 229. | Are the loading / unloading procedures in place and safety precautions displayed? | Loading / unloading procedures in place and safety precautions displayed. | 123. Loading / unloading procedures to be displayed. |
| 230. | Is there a provision to check the healthiness of road tanker with respect to explosives rules? | Checklist available for check the healthiness of road tanker with respect to explosives rules. | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| 231. | Are loaded tankers or trucks parked in a specific area on-site? | Loaded tankers or trucks parked in a specific area on-site. | --- |
| 232. | Do all truck and tanker drivers carry transport emergency (TREM) card or instruction booklet? | Truck and tanker drivers carried out transport emergency (TREM) card or instruction booklet. | --- |
| 233. | Do all truck and tanker drivers get training in handling emergencies during transport? | Tanker driver carrying hazardous materials were trained by approved training institute. | 124. Ensure that all truck and tanker drivers trained in handling emergencies during transport (As per rule 9 of MV Act). |
| 234. | Are all the tankers marked for proper Hazchem code? | Tankers marked for proper Hazchem code. | --- |
| | **2) Rail** | | |
| 235. | What hazardous materials are transported by rail? | No Hazardous materials transported by rail. | --- |
| 236. | Does the company have a direct siding on site? | Not Applicable | --- |
| 237. | Are tankers or other wagons used in transportation? | Not Applicable | --- |
| | **3) Pipelines** | | |
| 238. | What materials are transported to and from the site by pipelines? | Natural Gas pipelines | --- |
| 239. | Are the pipelines underground or over ground? | Pipelines were above ground | --- |
| 240. | Are corrosion protection measures employed in pipelines? | Yes, regular thickness test and painting done | --- |
| 241. | Whether intermediate booster pumps are used? | No | --- |
| 242. | What is the maximum, minimum and average transfer rates? | Details available with concern department | --- |
| 243. | Are the pipelines extended in the public domain? | Pipelines were not extended in the public domain | --- |
| 244. | Are the pipelines dedicated for each type of chemicals? | Yes, pipelines dedicated for each type of chemicals | --- |
| 245. | Are the pipelines fitted with safety equipment such as leak detectors, automatic | Manual shut-off valve was available. | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | shut-off valves etc.? | | |
| 246. | What is the frequency and method of testing of the pipeline? | Records available with maintenance Department. | --- |
| 247. | Is there written procedure for tackling leakages in pipeline? | Not evident | --- |
| | **C-20.2 Handling of Hazardous Substances** | | |
| 248. | What are the hazardous substances handled in the factory? | List of Hazardous material. Kindly see para 6.2 "List of Hazardous Raw Materials". (Isobutylene, n-Hexane, Toluene, Acrylonitrile, Triethyl Amine, Ethanol, IPA, EDC, Methanol, Benzaldehyde, DMF, MEG, Phenol, Carbon Tetrachloride, Bromine, Chlorine, Caustic Lye, HCl, Sulphuric Acid, Spent Acid, Formic Acid, TC, NSS, C-9, O-Xylene, MDC, Benzene, Ethyl Acetate, Acetic Acid, Mandelo Nitrile, Hydrazine Hydride, Aluminium Chloride, Sodium Cyanide, Acetonitrile, DMA, ODCB, 2,5 DCA, $SO_2$, $CO_2$, Methyl Chloride, HSD) | --- |
| 249. | Whether quantity of hazardous substances is above the threshold limit specified in the Manufacture, Storage and Handling of Hazardous Substances Rule, 1989? If yes, then required documentation is available as per the rule. | Yes, On site emergency plan, safety audit report, etc. available, as per para 6.2<br><br>Necessary License was evident.<br>For Details kindly refer "Annexure-1 List of Licenses/ Certification and its validity". | --- |
| 250. | Whether written procedure for handling the hazardous substance is available and operators are trained for handling such substances including actions required in case of leakages and spillages? | Work Instructions were prepared for safe handling of chemicals / substances. | 125. Periodic safety training on safe handling of chemicals should be imparted to workers by DISH approved training institute. |
| 251. | Are the employees aware of the hazards arising from hazardous substances and safety precautions to be | Most of Employees aware of the hazards arising from hazardous substances and safety precautions taken | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | taken during handling of these? | during handling of hazardous substances. | |
| | **C-20.3 Material Safety Data Sheets (MSDS)** | | |
| 252. | Are the material safety data sheets available for all the chemicals handled, used and manufactured in the factory? | Material safety data sheets (MSDS) available for the chemicals handled, used and manufactured in the factory. | --- |
| 253. | Whether the latest MSDS are displayed at strategic locations? | MSDS abstract were displayed at strategic locations. | --- |
| 254. | Is it available in local language? | MSDS abstract available in English only. | 126. To be made available in local language. |
| | **C-20.4 Spill Control Measures** | | |
| 255. | Whether spill control procedure is available? | Spill control procedure was available. | 127. Spill control procedure to be displayed. |
| 256. | Whether spill collection pit / sump is available at the workplace? | Spill collection pit / sump was available at the workplace. | --- |
| 257. | Whether methodology for recovery / disposal of collected material has been established? | Methodology for recovery / disposal of collected material has been established. | --- |
| | **C-20.5 Storage of Hazardous Substances** | | |
| 258. | Whether storage vessels are identified with the capacity as required under MSIHC, Rules 1989. | List of Hazardous material. Kindly see para 6.2 "List of Hazardous Raw Materials". (Isobutylene, n-Hexane, Toluene, Acrylonitrile, Triethyl Amine, Ethanol, IPA, EDC, Methanol, Benzaldehyde, DMF, MEG, Phenol, Carbon Tetrachloride, Bromine, Chlorine, Caustic Lye, HCl, Sulphuric Acid, Spent Acid, Formic Acid, TC, NSS, C-9, O-Xylene, MDC, Benzene, Ethyl Acetate, Acetic Acid, Mandelo Nitrile, Hydrazine Hydride, Aluminium Chloride, Sodium Cyanide, Acetonitrile, DMA, ODCB, 2,5 DCA, $SO_2$, $CO_2$, Methyl Chloride, HSD) | --- |
| 259. | What are the storage pressure and temperature? | Kindly see para 6.2 "List of Hazardous Raw Materials". For storage pressure and temperature. | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| 260. | Whether vessels are above ground / underground? | Vessels were above ground as well as underground. | --- |
| 261. | If any of the tanks storing flammable material, whether electrical equipment and fittings are as per electrical area classification? | The tanks storing flammable material & electrical equipment and fittings were flameproof type. | 128. Ensure that the electrical fittings are as per area classification. |
| 262. | Is the bunded area takes into account the total quantity of the largest tank? | Adequate bunded area observed. | --- |
| 263. | Whether the bund perimeter takes into consideration of trajectory of leak from tank? | The bund perimeter takes into consideration of trajectory of leak from tank. Dyke Guideline available. | --- |
| 264. | Are the vessels properly bonded and earthed and whether periodically checked and record maintained? | The vessels properly bonded and earthed and regularly checked. | --- |
| 265. | Are the vessels fitted with remotely controlled isolation valves? | Vessels were not fitted with remotely controlled isolation valves. | --- |
| 266. | Are vessels provided with emergency vent, relief valve, bursting disc, level indicator, pressure gauge, overflow line? | As applicable, installed. Data available with concerned plant / department. | --- |
| 267. | Where do such vents discharge? | Scrubbers available | --- |
| 268. | Are the vessels provided with alarms for high level, high temperature and high pressure? | Vessels were provided with alarm for high level, high temperature and high pressure. | --- |
| 269. | Are standby empty tanks or any other alternate systems provided for emptying / transfer in case of emergencies? | Standby empty tank was available. | --- |
| 270. | What are the provisions made for firefighting / tackling emergency situations around the storage vessels? | Following provisions made for firefighting / tackling emergency situations.<br>• Fire hydrant System<br>• Fire Extinguishers<br>• Sprinkler<br>• Sand Buckets<br>• SCBA Sets | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| 271. | Has any consequence analysis for loss of containment been carried out? | Consequence analysis (QRA study) for loss of containment was carried out. | --- |
| 272. | Whether the vessels are tested as per statute? | The vessels were tested as per statute. | --- |
| 273. | Whether log sheets are filled up on daily basis for recording the parameters of these vessels? | Log sheets were filled up on daily basis for recording the parameters of these vessels. | --- |
| 274. | Whether monitors for detection of leakage of flammable / toxic material installed? | $Br_2$ detector, $Cl_2$ detector, HCN detector, LEL detector, VOC meter & Portable LEL detector, $O_2$, Multi gas detector, etc. available | --- |
| 275. | Whether the chemicals stored are as per their compatibility? | Chemicals stored as per their chemical compatibility chart. | --- |
| | **C-20.6 Gas Cylinders** | | |
| 276. | What are the various gas cylinders used in the plant? | $Cl_2$, Nitrogen, $H_2$, Ammonia, zero air, $O_2$, DA, etc. | --- |
| 277. | Are valid licenses available for storing all these cylinders? | License available for $Cl_2$ storage. | --- |
| 278. | Are the cylinders stored and segregated as per their compatibility? | The cylinders stored and segregated as per their compatibility. | --- |
| 279. | What are the measures taken for combating any emergency in the cylinder's storage area? | Fire extinguishers were available for combating any emergency in the cylinder's storage area. | --- |
| 280. | Whether integrity test certificates are obtained from the suppliers of the cylinders? | Its informed that integrity test certificates were obtained from the suppliers of the cylinders. | --- |
| 281. | Are the cylinders chained and secured properly along with the valve caps and proper identification colour code? | Most of cylinders chained and secured properly along with the valve caps. | --- |
| 282. | Are the cylinders protected from heat or sun and rain? | Shed provided for the cylinder protection from heat or sun and rain. | --- |
| 283. | Whether monitors for detection of leakage of | $Br_2$ detector, $Cl_2$ detector, HCN detector, LEL | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | flammable / toxic gas installed? | detector, VOC meter & Portable LEL detector, $O_2$, Multi gas detector, etc. available | |
| | **C-20.7 Labelling and Colour Coding** | | |
| 284. | Are all the containers, vessels and storage tanks labelled for its content and capacity? | The containers, vessels and storage tanks labelled for its content and capacity. | --- |
| 285. | Whether the pipelines are colour coded as per IS 2379? | The pipelines were colour coded as per IS 2379. | 129. Ensure that pipelines are colour coded as per IS 2379. |
| 286. | Is any plant specific colour code followed? | Plant specific colour code was followed. | --- |
| 287. | Whether the colour codes are displayed conspicuously in the working areas? | The colour codes were not displayed conspicuously in the working areas. | 130. Colour code to be displayed conspicuously in the working area. |
| | **C-20.8 Hazardous Waste Management** | | |
| 288. | Is identification done for various types of hazardous wastes? | Identification done for various types of hazardous wastes as per GPCB Norms. | --- |
| 289. | Are these quantities less than those specified by the Hazardous Wastes (Management & Handling) Rules, 1989? | We came to know that quantities less than those specified by the Hazardous Wastes (Management & Handling) Rules, 1989. | --- |
| 290. | What are their disposal modes? | Their disposal modes as per GPCB Norms.<br>Waste disposal permission was evident. | --- |
| 291. | What are the systems / measures adopted for controlling air / water / land pollution? | The systems / measures adopted for controlling air / water / land pollution as per GPCB Norms. | --- |
| 292. | Whether the solid waste like combustibles, plastic, metals etc. segregated? | The solid waste like combustibles, plastic, metals etc. were segregated. | --- |
| | **C-21 Fire and Explosion Hazard** | | |
| | **C-21.1 Organizational Set-up for Fire Fighting** | | |
| 293. | What is the total strength of fire station and fire crew? | 2 firemen available | --- |
| 294. | How many fire crews are available in each shift? | 1 firemen & 1 safety officer was available in each shift. | --- |
| 295. | Is there fire squad identified | Fire-fighting trained | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | in each shift? | persons were available in each shift and fire squad identified with sticker on helmet. | |
| 296. | Standing fire order is available with latest revision | Standing Fire order was available with safety officer. | --- |
| 297. | How is the communication with fire station? | Communication with external fire brigade by telephone / mobile. | --- |
| 298. | Does fire safety inspections carried out? | Fire safety inspections carried out by safety officer | --- |
| 299. | Does emergency procedure available for leakage or combustion of flammables? | Emergency procedure available for leakage or combustion of flammables. | --- |
| 300. | What measures are available to control the fire load in the plant area? | Fire extinguisher, Fire hydrant, sprinkler, etc. available to control the fire load in the plant area. | --- |
| 301. | Whether technical knowledge and skills of the manager and staff responsible for overall fire safety of the plant is adequate? | Technical knowledge and skills of the manager and staff responsible for overall fire safety of the plant was adequate. | --- |
| 302. | How many major and minor incidents / fires were there in the factory during the last five years? Give department / plant wise. | Data available with EHS department. | --- |
| 303. | Have all the fires / incidents been investigated and corrective actions taken? Give break-up. | Fires / incidents been investigated and corrective actions taken.<br>Data was available at EHS department. | --- |
| 304. | **Resources:**<br>1. Adequacy of protective clothing (coat, trouser, gloves, boots and helmets);<br>2. Availability of SCBA for firefighting operations and spare cylinders (at least 2 for each SCBA);<br>3. Adequacy of hose, nozzles, ladders, | 1. Adequate<br>2. SCBA Set available<br>3. Adequate<br>4. Adequate | 131. To be make sure that, spare cylinders (at least 2 for each SCBA Set) to be provided. |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | lighting equipment and pumps; and<br>4. Communication facility at fire station, walkie talkie sets during firefighting. | | |
| | **C-21.2 Built in Safety in Civil Design and Construction** | | |
| 305. | Whether the two-safe means of escape available? Are they in separate directions? | Two safe means of escape available in separate directions. | --- |
| 306. | Is emergency exits provided to the building handling flammables? | Emergency exits provided to the building handling flammables. | --- |
| 307. | Whether emergency lights are provided? | Emergency lights were provided. | --- |
| 308. | Whether fire / smoke detectors are installed in fire prone areas? | Gas detectors were installed at some areas. | --- |
| 309. | Whether fire call points are provided in different areas? | Fire call points provided in different areas. | --- |
| 310. | Whether Fire hydrants are provided near the buildings? | Fire hydrants points were provided near the buildings. | --- |
| 311. | Is ventilation system in plant handling flammables is adequate to prevent formation of flammable mixtures? | Ventilation was adequate to prevent formation of flammable mixtures. | --- |
| 312. | Is adequate separation is provided between combustible / flammable materials and other material to restrict the fire growth? | Adequate separation was provided between combustible / flammable materials and other material to restrict the fire growth. | --- |
| 313. | Access routes for firefighting operations is available for areas having high fire load | Access routes for firefighting operations was available for areas having high fire load. | --- |
| 314. | Whether building changes interferes with fire detection and / or fire suppression systems? | No changes made in building after installation of Fire Suppression System. | --- |
| 315. | Whether building changes cause unreasonable fire loading / openings in the fire rated walls? | No fire rated walls available. | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | **C-21.3 Built in Safety in Electric Circuits and Equipment** | | |
| 316. | Are the electrical equipment in areas where flammables mixture is likely to be present of flame-proof type? | Flameproof type electrical installations were provided. | --- |
| 317. | Are lightning arrestors are provided to the buildings / structures storing flammable materials? | Lightning Arrestors were provided to the buildings / structures storing flammable materials. | 132. Ensure all plant and other buildings, tall structures are protected from lightning. |
| 318. | Whether adequate bonding and grounding of electrical equipment / pipelines provided? | Bonding and grounding were provided in electrical equipment / pipelines. | --- |
| | **C-21.4 Explosive Substances (Explosive Act ,1884 and the Explosive Rules,1983)** | | |
| 319. | Whether necessary license / approval taken from concerned statutory bodies? | No explosive substance used in factory. | --- |
| 320. | Whether systems for explosion suppression, high speed fire detection with deluge, sprinklers, explosion venting etc. are provided? | Not Applicable | --- |
| 321. | Whether explosion resistant walls or barricades are provided around explosive storage? | Not Applicable | --- |
| 322. | Whether explosive substance storage areas are restricted for entry? | Not Applicable | --- |
| 323. | Whether only trained persons are handling explosive substances? | Not Applicable | --- |
| 324. | Whether explosive substances are stored and transported in approved containers only? | Not Applicable | --- |
| 325. | Whether electrical fixtures in areas handling explosives are explosion proof type? | Not Applicable | --- |
| 326. | Whether adequate measures are taken to prevent any sources of | Not Applicable | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | ignition where explosive substances are handled? | | |
| | **C-21.5 Fire Safety in Handling Flammable and Explosive materials** | | |
| 327. | Whether emergency procedure is available for control of leakage? | Emergency procedure was available for control of leakage. | --- |
| 328. | Whether emergency measures are displayed locally in case of accidental spillage / leakage? | Dos & Don'ts displayed at some area | --- |
| 329. | Whether facility is provided for safe drainage of combustible or flammable liquids in case of leakages? | Dyke was provided for safe drainage of chemicals in case of leakages. | --- |
| 330. | Whether highly flammable liquids are stored under inert atmosphere? | Highly flammable liquid stored as per PESO guideline. | --- |
| 331. | Whether flammable storage tanks are provided with flame arrestors? | Flammable storage tanks were provided with flame arrestors. | --- |
| 332. | Whether suitable PPEs are provided? | Suitable PPEs were provided. | --- |
| | **C-21.6 Fire Detection and Alarm System** | | |
| 333. | What type of fire detection and alarm system provided? | Fire detection & alarm system were not seen during audit. | 133. Suitable fire detection & alarm system to be installed wherever necessary. |
| 334. | Whether all fire prone areas of the plant are covered with fire detection system? | Fire detection & alarm system were not seen during audit. | --- |
| 335. | Whether fire detection equipment and smoke alarms in good operating condition? | Fire detection & alarm system were not seen during audit. | --- |
| 336. | Whether the number of fire call points are adequate and free from obstruction? | Fire call point was not seen. | --- |
| 337. | Whether regular inspection / maintenance / testing of fire detection and alarm system carried out and records maintained | Fire detection & alarm system were not seen during audit. | --- |
| 338. | Whether any atmospheric monitoring is carried out for explosive mixture of gases or vapours? | Workplace monitoring was carried out by third party. | 134. Workplace monitoring to be carried out at regular intervals and |

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| | | | records to be maintained. |
| 339. | Whether emergency power supplies are provided to fire detection and fire alarm system? | DG set available for emergency power supplies. | --- |
| 340. | Whether smoke detectors are located considering ventilation pattern? | Fire detection & alarm system were not seen during audit. | --- |
| 341. | Whether annunciation of fire is local or in the control room or in both places? | Fire detection & alarm system were not seen during audit. | --- |
| 342. | Whether fire panel is constantly attended? | Fire detection & alarm system were not seen during audit. | --- |
| | **C-21.7 Passive and Active Fire Protection System** | | |
| 343. | What are the passive fire protection measures available? (Barriers, doors, dampers etc.) | Passive fire protection was not evident during audit. | 135. Passive fire protection measures to be provided wherever required. |
| 344. | Are the areas requiring fire barriers identified? | Area was identified. | --- |
| 345. | Whether the fire barrier provided is of adequate ratings? | Fire barrier was not provided. | --- |
| 346. | Whether ventilation ducts in flammable areas have been provided with isolation dampers of suitable fire rating? | No Ventilation ducts was provided in flammable area. | --- |
| 347. | Whether sprinklers / deluge are installed wherever necessary? | Sprinkler system was installed in IB Storage Area, Solvent Storage Area, Bromine Storage Area, Ammonia Storage area, Coal Yard, etc. | --- |
| 348. | Whether regular inspection / maintenance / testing of fire protection system carried out and records maintained? | Regular inspection of fire protection system was carried out and records were maintained. | --- |
| | **C-21.8 Fixed Fire Extinguishing System** | | |
| 349. | What are the sources of firewater and whether they are dedicated to the fire extinguishing system? | GIDC water was the source of firewater (underground tank) and they were dedicated to the fire extinguishing system. | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| 350. | Whether the capacity of dedicated water reservoir is adequate to supply to hydrants for minimum 2 h? | The capacity of dedicated water tank was adequate to supply to hydrant for minimum 2 hrs. Fire Water Capacity: 600 KL | --- |
| 351. | Whether un-interrupted power supply is provided to the firewater pumps? | Un-interrupted power supply is provided to fire water pumps. | --- |
| 352. | Whether the extinguishing medium selected is appropriate to the class of fire (water, gaseous, foam, dry powder)? | The extinguishing medium selected was appropriate to the class of fire. | --- |
| 353. | Whether fire hydrants layout is available? | Yes, Fire hydrants layout was available. | --- |
| 354. | Whether additional (over minimum requirement) fire hoses, nozzles are available? | Additional (over minimum requirement) fire hoses, nozzles were available. | --- |
| 355. | Whether the hydrants lines are kept pressurized? | Yes, hydrants lines were kept pressurized.<br>Line pressure – 7 kg/cm$^2$ | --- |
| 356. | Whether regular inspection / maintenance / testing of fixed fire extinguishing systems carried out and records maintained? | Regular inspection of fixed fire extinguishing systems was carried out by internally and records were maintained | --- |
| | **C-21.9 Portable Fire Extinguishing System** | | |
| 357. | Whether suitable type and numbers of fire extinguishers provided? | Total 351 Nos. of Fire Extinguishers installed in entire premises. | --- |
| 358. | Whether the fire extinguishers are located at conspicuous position and easily accessible? Are they fully charged and tagged? | The fire extinguishers were located at conspicuous position and easily accessible & they fully charged and tagged. | --- |
| 359. | Whether fire extinguishers periodically inspected, tested, refilled and records maintained? | Fire extinguishers were periodically inspected, tested and refilled and its records was also available. (Fire extinguisher: - Every quarter by externally & Every month done by Internally). | --- |
| 360. | Whether defective / unchecked fire extinguishers present at site? | Defective fire extinguishers were not present at site. | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| 361. | Whether additional fire extinguishers are available? | Additional fire extinguishers were available. | --- |
| | **C-21.10 Fire Fighting Equipment and Facilities** | | |
| 362. | Whether fire tenders (water / foam) are available? | Fire tender was not available. | --- |
| 363. | Whether the fire-fighting system and equipment approved, tested and maintained as per relevant standard? | The fire-fighting system and equipment approved, tested and maintained as per relevant standard. | --- |
| 364. | Whether the SCBA / fire suit provided to firefighting team for fire action? | SCBA sets (10 Nos. including spare set) were provided to firefighting team for fire action. | 136. To be make sure that, spare cylinders (at least 2 for each SCBA Set) to be provided. |
| 365. | What is system for maintenance/ recharge of SCBA? | SCBA set inspection, maintenance, cylinder testing and recharge was carried out. | --- |
| 366. | Is proper access available for fire-fighting equipment? | Proper access available for fire-fighting equipment. | --- |
| 367. | Whether fire hose cabinets are in good condition, easily visible, and accessible? | Fire hose cabinets were in good condition, easily visible and accessible. | --- |
| 368. | Whether drill tower is available? Are fire personnel carrying out regular fire drill? | Drill tower was not available.<br>Fire Drill / Firefighting training was carried out regular interval. | --- |
| 369. | What is the communication facility at fire station? Is it adequate? | Communication facility at fire station: Telephone / mobile | --- |
| | **C-21.11 Fire Drill** | | |
| 370. | Whether mock fire drills are conducted? What is the frequency of drills? | Mock Drills conducted on quarterly basis.<br>Last drill – 28/03/2024 | 137. Firefighting drill to be carried out at every 2 years as per rule 66-A of the GFR-1963. |
| 371. | Whether fire drills are also performed in night Shift? | Its informed that Fire drill will be performed in night Shift. | --- |
| 372. | Whether feedback of fire drill is documented? | Mock drill feedback records was documented. | --- |
| 373. | What is the system of mutual-aid scheme? | Mutual aid agreements with nearby fire stations & industries | --- |
| | **C-21.12 Fire Fighting Training** | | |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| 374. | Whether there is a system of providing fire- fighting training to plant personnel? | Yes, system of providing firefighting training to plant personnel was available. | --- |
| 375. | What is the frequency and duration of such training? Whether training records are maintained? | At regular interval (30 mins), firefighting training was provided and its record maintained. | --- |
| 376. | Whether fire squads are identified for different areas for first-aid firefighting and rescue, and suitably trained? | Fire-fighting trained persons were available and fire squad was identified with sticker on helmet & first-aid firefighting and rescue suitably trained. | --- |
| 377. | Are all personnel conversant with the fire prevention and protection measures? | Most of personnel were familiar with the fire prevention and protection measures to be taken. | --- |
| 378. | Whether the fire staff are sent for refresher / advanced training courses? | Firefighting training was given internally. | 138. Fire staff to be sent for advanced training courses. |
| | **C-21.13 Static Electricity and Lightning** | | |
| 379. | Whether all vessels and pipes are provided with suitable bonding and grounding? | Most of the vessels and pipes were provided with suitable bonding and grounding. | 139. Double earthing to be ensured. |
| 380. | Whether arrangement has been made for grounding the tanker containing flammable liquid during loading / unloading? | Earthing available for grounding the tanker containing flammable liquid during loading / unloading. | --- |
| 381. | Whether spark resistant tools are provided? | Spark resistant tools were provided as & when required. | --- |
| 382. | Whether lightning protection is provided and is adequate? | Lightning Arrestors installed. | 140. Make sure that installed lightning arrestors are adequate. |
| 383. | Whether antistatic clothing, hand gloves and footwear are provided? | Antistatic clothing, hand gloves and footwear were provided as and when required. | --- |
| | **C-21.14 Pressure Relief System (The FA 1948 section 31, The GFR – Rule 61)** | | |
| 384. | Whether the listing of all ‘pressure plants’ [as defined under Factories Act] has been done? | Pressure plant area identified and pressure plant list maintained by concern department. | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | **C-22 Industrial Hygiene / Occupational Health** | | |
| | **C-22.1 Ventilation, Illumination, Noise, Vibration, Heat stress and Non-ionizing Radiations** | | |
| | **C-22.1.1 Ventilation** | | |
| 385. | Whether any ventilation study has been carried out? | Ventilation study was not evident. Natural ventilation was adequate in plant. | --- |
| 386. | Whether natural ventilation is adequate or not? | It was informed that natural ventilation was adequate in plant. | --- |
| 387. | Whether dust / fumes / hot air is generated in the process? | Dust / Fumes generated in the process. | --- |
| 388. | Is there any exhaust ventilation system in any section of the plant? | Natural ventilation system in plant. | --- |
| 389. | Is periodic / preventive maintenance of ventilation system carried out and record is maintained? | As natural ventilation so, no such records was evident. | --- |
| 390. | Does any ventilation system re-circulate the exhausted air in work areas? | Natural ventilation system in plant. | --- |
| 391. | Is the work environment assessed and monitored for chemical and physical hazards? | The work environment assessed and monitored for chemical and physical hazards.<br>Form no. 37 monitoring reports was available. | 141. Work place monitoring to be carried out at regular intervals and records to be maintained. |
| 392. | Whether PPE are provided to workers exposed to dust / fumes and gases? | Adequate PPEs provided to workers | --- |
| | **C-22.1.2 Illumination** | | |
| 393. | Whether illumination study has been carried out for the assessment of illumination level? | Illumination study was carried out for the assessment of illumination level. | --- |
| 394. | Is there any system of periodical cleaning and replacing the light fittings / lamps in order to ensure that they give the intended illumination levels? | System available for periodical cleaning and replacing the light fittings / lamps in order to ensure that they give the intended illumination levels. | --- |
| 395. | Are the workers subject to periodic optometry tests and records maintained? | Periodic optometry test was carried out for the workers during periodical | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | | medical check-ups. | |
| 396. | Are emergency lighting available at first aid centre? | Emergency light was available in OHC. | --- |
| | **C-22.1.3 Noise (The GFR 102 Schedule 23)** | | |
| 397. | Whether any noise study conducted? | Noise study conducted by third party & records maintained. | --- |
| 398. | Are there any machines / processes generating high noise? | High noise area i.e., Compressor, Blower, Boiler, etc. | --- |
| 399. | Whether engineering and administrative controls been implemented to reduce noise exposure below the permissible limits? | Engineering and Administrative Control: Noise monitoring, PPE's, training etc. | --- |
| 400. | Is there a system of subjecting all those employees to periodic audiometric test who work in high- level noise areas? | Periodic Audiometric test was carried out for high level noise area. | --- |
| 401. | Whether the workers are made aware of the ill effects of high noise? | Workers were aware of ill effects of high noise. | --- |
| 402. | Whether earmuffs / plugs are provided and used? | Earmuffs / plugs were provided and used. | --- |
| | **C-22.1.4 Vibration (The GFR 102 schedule 23)** | | |
| 403. | Are there equipment which contributes excess level of vibrations and whether they are identified? | Equipment identified which generate excess level of vibrations. | --- |
| 404. | Whether any vibration study has been carried out? | Vibrations study was not evident. During maintenance, vibration measures. | --- |
| 405. | Are the measures taken to combat vibration to acceptable levels? | Adequate measures taken to combat vibration to acceptable levels. | --- |
| 406. | What is the frequency for measurements of vibration? | During preventive maintenance, vibration measures. | --- |
| 407. | Are the records of measurements and maintenance of equipment / system maintained? | Records of maintenance of equipment / system maintained at concern department. | --- |
| | **C-22.1.5 Heat Stress / Cold stress (Extremes of Temperature)** | | |
| 408. | Are there sources from | Sources from equipment | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | equipment increasing the heat load in workplaces? | increasing the heat load in work places: Boiler | |
| 409. | Whether evaluation of heat stress is carried out? | Heat Stress evaluation was not carried out | --- |
| 410. | Whether natural ventilation is adequate to minimize the heat stress in work environment? | Natural ventilation was adequate to minimize the heat stress in work environment. | --- |
| 411. | Are resources available to deal with very hot or very cold conditions (drinking water, lined gloves, insulated boots)? | Appropriate PPEs, cold drinking water were provided.<br>Proper engineering control & Administrative control made available | --- |
| 412. | Do workers know the symptoms of heat cramps / heatstroke or frost bite / hypothermia? | Workers know the symptoms of heat cramps / heatstroke. | --- |
| 413. | Are the personal protective equipment suitable for reducing the effects of heat stress available? | The personal protective equipment suitable for reducing the effects of heat stress available | --- |
| | **C-22.1.6 Non-ionizing Radiations** | | |
| 414. | Does the work involve likely exposure to non-ionising radiations (ultraviolet, infrared, radiofrequency, microwaves, lasers, etc.) | Non-ionizing radiations source was not available. | --- |
| 415. | Whether risk assessment have been done for all work areas involving presence of non-ionising radiations? | Not Applicable | --- |
| 416. | Are the work areas displayed with relevant safety signs? | Not Applicable | --- |
| 417. | Are the employees aware about the hazards of non-ionising radiations? | Not Applicable | --- |
| 418. | Do written procedures exists for working in non-ionising radiations? | Not Applicable | --- |
| 419. | Is the work environment monitored periodically for physical hazards and control measures initiated whenever deviation from | Not Applicable | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | permissible values is observed? | | |
| 420. | Whether suitable personal protective equipment are provided to workers exposed to non-ionising radiations? | Not Applicable | --- |
| | **C-22.2 Workplace Monitoring for Hazardous Chemicals (Sec. 7-A FA 1948 & Rule 12-B of GFR, 1963)** | | |
| 421. | Whether the dust, fumes, smoke, aerosols and mist are monitored as per statute and records maintained? | Workplace monitoring was carried out and records maintained in form-37. | 142. Make sure that workplace monitoring to be carried out at regular intervals. |
| 422. | What are the types of detectors used for monitoring concentration of hazardous chemicals? | $Br_2$ detector, $Cl_2$ detector, HCN detector, LEL detector, VOC meter & Portable LEL detector, $O_2$, Multi gas detector, etc. available | --- |
| 423. | Is any alarm system installed for any leakage of hazardous chemicals? | Alarm system provided for any leakage of hazardous chemicals ($Br_2$ detector, $Cl_2$ detector, HCN detector, LEL detector, VOC meter & Portable LEL detector, $O_2$, Multi gas detector, etc.) | --- |
| 424. | Are antidotes available for toxic chemicals? | Necessary antidotes were available at OHC. | --- |
| 425. | Are control measures initiated whenever deviation from permissible values is observed? | Adequate control measures provided whenever deviation from permissible values is observed. | --- |
| | **C-22.3 First Aid Facilities and Occupational Health Centre (OHC) (The FA Sec.-45, Gujarat Factories Rules, 1963-Rule 68-U, 68-V, 71)** | | |
| 426. | Are adequate numbers of first aid boxes provided? Give location details? | First Aid Box: 15 Nos. List available at OHC & EHS Dept. | --- |
| 427. | Are qualified / trained first aiders available in each shift? | Total First Aider: 10 Nos. Qualified & trained first aiders available in each shift. | --- |
| 428. | How many qualified / trained first aiders are available at each plant / department? | Total 10 Nos. of qualified & trained first aiders available. | --- |
| 429. | How many persons are | Total 10 Nos. of person | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | trained / given refreshers training in first aid in a year? | trained in first aid training and data maintained. | |
| 430. | Whether occupational health centre is provided? | Occupational health centre was available. | --- |
| 431. | Does OHC conform to the provisions of the existing statutes? | OHC was conform the provision of the statues. | 143. OHC to be reviewed as per rule 68-U of GFR-1963. |
| 432. | Are the Medical Attendants / Doctors available in each shift? | Medical attendants were available in the OHC.<br>FMO - Dr. Tejas Mehta<br>Total first aider - 10 Nos. | --- |
| 433. | What facilities are available for transportation of the injured to hospital? | Dedicated Vehicle was available for transportation of the injured to hospital. | --- |
| 434. | Are the names of the trained first aiders displayed? | The notice containing names and telephone number of first aider was displayed at main gate | --- |
| 435. | Are the name of nearest hospitals and its telephone number available in OHC? | Name of nearest hospitals and its telephone number was available in OHC & Security Gate. | --- |
| 436. | Does the plant have any special preventive medicine program? | Plant have not conducted special preventive medicine program | --- |
| 437. | Is ambulance posted in proper place and is it available whenever required? | Dedicated vehicle was available. | --- |
| 438. | Are sufficient numbers of anti-dotes available in case of any emergency? | Necessary antidotes were available at OHC. | --- |
| 439. | Are fire safety measures provided in first aid centre? | Fire safety measures were provided nearby. | --- |
| 440. | Are emergency lighting arrangements available at first aid centre? | Emergency light was available in OHC. | --- |
| | **C-22.4 Periodic Medical Examination (The GFR Rule 68-T, Rule 102)** | | |
| 441. | Whether the periodical medical examination of employees, required under relevant statute is carried out? | The periodical medical examination (Form No. 32) of employees, required under relevant statute was carried out. | --- |
| 442. | Whether it is ensured that contractors employee are medically examined during pre-employment as well as | It was informed that contractors employee were medically examined during pre-employment as well as | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | during the course of employment? | during the course of employment. | |
| 443. | During the periodical medical examination of the workers, are they examined as per the hazardous process in which they work? (First schedule of The Factories Act, 1948) | During the periodical medical examination of the workers, they were examined as per the hazardous process in which they work. (First schedule of The Factories Act, 1948) | --- |
| 444. | Are the records of all such examination maintained? | Records of all such examination was maintained. | --- |
| | **C-22.5 Personal Protective Equipment and Emergency Equipment** | | |
| 445. | Whether list of required PPE for each hazardous activity is available? | Hazard Activity chart with PPE (PPE matrix available for each plant) was available. | --- |
| 446. | Whether feedback from workers obtained during selection of PPE? | Feedback of workers was obtained during selection of PPE. | --- |
| 447. | Have the workers been trained in proper use of PPE including BA sets? | Training on use of PPE, SCBA set was imparted to workers and records were maintained. | --- |
| 448. | What is the system of procurement, inspection, issue, maintenance and replacement of PPE? | System was available for procurement, inspection, issue, maintenance and replacement of PPE. | --- |
| 449. | Whether qualitative and quantitative fit-check for respirators is ensured prior to use? | Qualitative and quantitative fit-check for respirators was carried out to prior to use. | --- |
| 450. | What are the arrangements for safe custody and storage of PPE? | Locker provided for safe custody and storage of PPE. | --- |
| 451. | Are the contractor's workers provided with the required PPE? | Contractor's workers provided with the required PPE. | --- |
| 452. | Do the PPE conform to any standard? | PPE conform by IS standard. | --- |
| 453. | Are sufficient eye wash fountains and safety showers available? | Eye wash fountains and safety showers were available in each plant. | --- |
| 454. | Whether appropriate respiratory protective devices are available in | Appropriate respiratory protective devices i.e., SCBA sets (10 Nos. | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | accordance to the hazard potential? | including spare set) were available in accordance to the hazard potential. | |
| 455. | Are the staff members trained in the right uses of respiratory protective devices? | Staff members trained in the right uses of respiratory protective devices. | --- |
| | **C-22.6 Occupational Diseases** | | |
| 456. | Whether pre-employment medical check-up data available? | Pre-employment medical check-up data available. (Form No. 33 available) | --- |
| 457. | During the medical check-up, is any person found having occupational diseases mentioned in 3rd schedule of The Factories Act, 1948? | During the medical check-up, any person not found having occupational diseases mentioned in 3rd schedule of The Factories Act, 1948. | --- |
| 458. | Whether the medical practitioner informed the Chief Inspector of Factories about the occurrence of the occupational disease? | No abnormality detected during periodic medical examination of workers. | --- |
| | **C-23 Accident / Incident Reporting, Investigation and Analysis (The GFR – Rule 103,111)** | | |
| | **C-23.1 Accident Reporting and Database Management** | | |
| 459. | What is the procedure for accident / incident /dangerous occurrence reporting? | Procedure for accident / incident / dangerous occurrence reporting was available. | --- |
| 460. | Whether the accident data for the last five years for reportable and non-reportable accidents are available? | Accident data for the last five years for reportable and non-reportable were available at EHS department. (Form - 29 maintained) | --- |
| | **C-23.2 Accident Investigation** | | |
| 461. | Are all the accidents investigated? | The accidents were investigated informally. | --- |
| 462. | Whether accident investigation procedure is documented? | Accident investigation procedure was documented. | --- |
| 463. | Whether accident investigation reports are submitted to top management? | Accident reports were submitted to top management. | --- |
| 464. | How are the findings from accident investigation | Learning from incident, tool box talk. | --- |

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|---|---|---|---|
| | reports communicated to workers? | | |
| | **C-23.3 Analysis of Accidents** | | |
| 465. | Whether accident analysis is done as per IS 3786? | Accident analysis was done informally. | 144. Accident analysis to be done as per IS 3786. |
| 466. | Whether root causes of accidents are analysed? | Root causes of accidents were analysed informally. | --- |
| 467. | Is the accident statistics effectively utilized? If yes, how? | The accident statistics was utilized. | --- |
| 468. | What nature of injuries occurred during the last five years? | Data available with EHS department. | --- |
| | **C-23.4 Implementation of Recommendations** | | |
| 469. | How does the management ensure implementation of the recommendations to avoid recurrence of accidents and incidents? | Recommendation reviewed in the review meetings and also tracking through system. | --- |
| | **C-23.5 Reporting and Investigation of Near miss Incidents** | | |
| 470. | Are all near-miss incidents reported and investigated? | We came to know that Near-miss incident were reported, investigated, and analysed. | 145. Impart training to all workers to observed near miss on-site. |
| 471. | Is there any system of classifying and analysing the near-miss incidents? | We came to know that system of classifying and analysing the near-miss incidents was available. | --- |
| | **C-24 EMERGENCY PREPAREDNESS** | | |
| | **C-24.1 Site Specific Details** | | |
| 472. | Are the site area maps (including layout, access roads and assembly points) available in control room / emergency control centre? | The site area maps (including layout, access roads and assembly points) was available. | --- |
| | **C-24.2 Duties and Responsibilities of Key Personnel** | | |
| 473. | Is the hierarchy of emergency response personnel right from site emergency controller downward, and alternative officials identified? | The hierarchy of emergency response personnel right from site emergency controller downward and alternative officials was identified.<br>(Ref.: - On-site emergency plan) | --- |
| 474. | Are the duties and | Duties and responsibilities | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | responsibilities assigned to the designated officials during emergency, both during and outside normal working hours clearly identified and understood by them? | assigned to the designated officials during emergency, both during and outside normal working hours clearly identified and understood by them.<br>(Ref.: On-site emergency plan) | |
| | **C-24.3 Identification of Emergencies and Accident Scenario** | | |
| 475. | Are the possible accident scenarios leading to emergency identified and known to the operating personnel? | All possible accident scenarios leading to emergency identified and known to the operating personnel.<br>(Ref.: - On-site emergency plan) | --- |
| 476. | Are approved emergency preparedness plans (on-site and off-site) in place? | Approved emergency preparedness plans (on-site emergency plan) was available (OSEP: August-2023). | --- |
| | **C-24.4 Declaration and Termination of Emergency** | | |
| 477. | Is the list of designated officials who are to be communicated about declaration and termination of emergency available in the control room / emergency control centre? | The list of designated officials who are to be communicated about declaration and termination of emergency available in the control room / emergency control centre.<br>(Ref.: On-site emergency plan) | --- |
| 478. | Are the methods of communication (siren, public address system etc.) for declaration and termination of an emergency known to all the workers? | Siren was available for declaration and termination of an emergency known to all the workers. | --- |
| | **C-24.5 Resources-evacuation / Transport** | | |
| 479. | Are the following resources (equipment, personnel and procedures) required to handle emergency available?<br>1. Communications,<br>2. Public announcement systems<br>3. Monitoring of | Following resources (equipment, personnel and procedures) available for handling emergency.<br>1. Siren, Landline Phone, etc.<br>2. Public Address System was not evident. | 146. PA System was not installed.<br>147. Direction for emergency exit / escape route should be marked in haulage / Alleyway. |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | hazardous releases into the environment,<br>4. Emergency shelters at the facility,<br>5. Emergency exits with proper illumination, with uninterrupted power supply,<br>6. Direction for emergency exit / escape route marked in haulage / Alleyways,<br>7. Transport for evacuation of plant personnel,<br>8. Medical care including administration of antidotes, and<br>9. Security / maintenance of law and order. | 3. Environment and Workplace monitoring carried out.<br>4. It was informed that emergency shelters available.<br>5. Emergency Exit provided and details available with EHS dept.<br>6. Assembly Points, Escape route, emergency exit marked.<br>7. Transport vehicle available.<br>8. For information see element C-9, C-22.3<br>9. Security staff available. | |
| | **C-24.6 Communication Facilities** | | |
| 480. | Does the emergency control centre have direct communication links with the fire station and the plant control room? | Landline phone, mobile, etc. available for emergency communication. | --- |
| 481. | Are there adequate alarm points from which an emergency alarm can be raised? | Adequate alarm points from which an emergency alarm can be raised. | --- |
| 482. | Is there infrastructure available for ensuring backup electric power supply for communication links where required | DG set available for ensuring backup electric power supply for communication links where required. | --- |
| | **C-24.7 Medical Care** | | |
| 483. | Is the procedure for emergency medical care available? | Emergency medical care (OHC) was available.<br>& Mutual aid with Jolva Hospital, Dahej. | --- |
| 484. | Whether the system has been tested at regular frequency through mock drill / exercises for its adequacy? | Quarterly mock drill was conducted in the factory.<br>Last mock drill – 28/03/2024.<br>Mock drill report was prepared. | --- |
| 485. | Does the system of periodic | System of periodic | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | replacement of antidotes and medicines required in emergency exist? | replacement of medicines required in emergency was evident. | |
| | **C-24.8 Updation of Emergency Plan** | | |
| 486. | Is the emergency plan updated based on the feedback from the periodic drills / exercises? | Emergency plan (On-site emergency plan) available and updated based on the feedback from the periodic drills. | 148. Ensure corrective actions for lacuna observed during mock drill is implemented. |
| 487. | Are the contact details of all concerned officials kept updated in the emergency plan? | Contact details of all concerned officials kept updated in the emergency plan. | --- |
| | **C-24.9 Periodic Drills / Exercises** | | |
| 488. | Are mock exercises conducted at stipulated intervals? | Quarterly mock drill was conducted in the factory.<br>Last mock drill – 28/03/2024. | --- |
| 489. | Are the scenarios varied in the mock-exercises to ensure that all possible factors including meteorological conditions, affected plant personnel covered? | Scenarios varied in the mock-exercises done that all possible factors including meteorological conditions, affected plant personnel covered.<br>Ref.: On-site Emergency plan | --- |
| 490. | Whether emergency preparedness Plans have been tested and reviewed at regular frequency through mock drill for its adequacy | Emergency preparedness Plans have been tested and reviewed at regular frequency through mock drill.<br>(Mock drill Duration: Quarterly) | --- |
| | **C-24.10 Training of Plant Personnel** | | |
| 491. | Are the plant personnel trained in handling emergency equipment? | Most of plant personnel trained in handling emergency equipment. | --- |
| | **C-24.11 Public Awareness Programme** | | |
| 492. | Are public awareness programs conducted for the people around the site regarding the actions to be taken in case of off-site emergency? | It was informed that Public Awareness programs conducted for the people around the site regarding the actions to be taken in case of off-site emergency. | --- |
| | **C-24.12 Mutual-aid Program** | | |
| 493. | Are the types of accidents where external | We came to know that all types of accidents where | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | organizations would be involved in remedial actions identified? Are their responsibilities defined? | external organizations would be involved in remedial actions were identified.<br>Their responsibilities were also defined in emergency plan. | |
| 494. | Is the plant responsible for rendering mutual aid assistance to any other external organizations? Does this assistance effect the plant's emergency preparedness? | Plant responsible for rendering mutual aid assistance to any other external organizations.<br>This assistance did not affect the plant's emergency preparedness | --- |
| 495. | Whether the communication channels for mutual assistance identified and known with and between two organizations? | The communication channels (Telephone / Mobile) for mutual assistance identified and known with and between two organizations | --- |
| | **C-24.13 Emergency Control Centre** | | |
| 496. | Is the emergency control centre located beyond the effective distances of identified emergency scenarios? | Emergency control centre located beyond the effective distances of identified emergency scenarios.<br>Four Assembly point evident. | --- |
| 497. | If the emergency control centre is located within the effect distance, is it suitably protected that it will be available in case of emergency? | The emergency control centre was not located within the effect distance; it is suitably protected that it will be available in case of emergency.<br>Four Assembly point evident. | --- |
| | **C-25 Safety Inspection** | | |
| | **C-25.1 Inspection Programme** | | |
| 498. | Are checklists available for inspections? For example, availability of checklists like:<br>a. Handling, Storage and Transportation of hazardous chemicals.<br>b. Electrical hazards;<br>c. Fire safety;<br>d. Hand and portable | We came to know that most of said Checklist was available. | 149. Ensure that all said checklist available and maintained. |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | power tools;<br>e. Machine hazards;<br>f. Lifting equipment;<br>g. Ladders and scaffolding;<br>h. Environmental Monitoring;<br>i. Civil structure;<br>j. Housekeeping;<br>k. Emergency equipment; and<br>l. Gas cylinder and other pressure vessels used /available in the organization. | | |
| | **C-25.2 Safety Related Deficiency (SRD) Report** | | |
| 499. | Are SRDs generated based on the area wise checklists? | Safety related deficiency was generated during plant round. | --- |
| 500. | What is the procedure for resolving the SRDs? | SOP & checklist was available. | --- |
| 501. | Whether the procedure exists for notification and root cause analysis of non-conformities and action taken on them? | Root cause analysis of non-conformities was carried out and action taken on them. | --- |
| | **C-25.3 Safety Inspection Records** | | |
| 502. | Are the safety inspection records maintained? | Safety inspection records were available with EHS department. | --- |
| | **C-25.4 Methodology and Inspection Team** | | |
| 503. | Is there written procedure for safety inspection? | Procedure exists for safety inspection. | --- |
| 504. | Whether safety inspection is carried out by a designated team? | Safety department was carried out safety inspection. | --- |
| 505. | What is the frequency of safety inspections? | Safety inspection carried out weekly & Monthly basis. | --- |
| 506. | Whether an inspection report is generated? | Inspection observations was recorded. | --- |
| | **C-25.5 Compliance of Recommendations** | | |
| 507. | To whom the recommendations are submitted | Recommendations were submitted to all HOD, Unit Head & Concern person. | --- |
| 508. | Are recommendations of safety inspections | We came to know that recommendations of | --- |

| Sr. No. | Checkpoints | Current Status / Observation | Recommendations |
|---|---|---|---|
| | complied in time? | safety inspections complied in time. | |
| 509. | Is compliance of recommendations sent to top management? | Compliance of recommendation sent to top management. | --- |
| 510. | Is compliance of recommendations reviewed by safety committee? | Compliance of recommendations reviewed by safety committee. | --- |
| 511. | Does top management follows-up the compliance? | Top management follows-up the compliance. | --- |
| 512. | Is compliance of recommendations sent to top management? | Compliance of recommendation sent to top management. | --- |

***END OF REPORT***